

Supplementary materials

Highly scalable technology assisted differential diagnostics of ASD

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S1 Version numbers for R and all packages

```
## [1] "R version 4.5.1 (2025-06-13)"
## [1] "tidyverse version 2.0.0"
## [1] "knitr version 1.50"
## [1] "ggpubr version 0.6.1"
## [1] "gggrain version 0.0.4"
## [1] "Hmisc version 5.2.3"
```

```
## [1] "rstatix version 0.7.2"
## [1] "emmeans version 1.11.2"
## [1] "flextable version 0.9.9"
## [1] "officer version 0.7.0"
## [1] "kableExtra version 1.4.0"
## [1] "english version 1.2.6"
```

S2 Data availability

Throughout the manuscript, we use a threshold of $\alpha = 0.05$ to assess significance. Data was preprocessed in R 4.3.0¹ using RStudio² and MATLAB R2023b³ with all machine learning analyses computed using Neurominer V1.3 curufin⁴. All scripts and preprocessed, anonymised data are available on https://github.com/IreneSophia/NEVIA_ASF-BPD/tree/main/ML_BPD-ASF, while full SVM models are provided on <https://osf.io/y8xbp>.

S3 Inclusion and exclusion criteria

Inclusion criteria applying to all participants:

- no neurological diagnoses
- at least 18 and at most 60 years of age
- normal or corrected-to-normal vision
- legally binding, written informed consent from the participant for study participation

We only included clinical participants with either a BPD or an ASD diagnosis but excluded anyone who was given both diagnoses.

Additional inclusion criteria for ASD participants:

- valid F84 diagnosis according to ICD-10
- normal speaking abilities

Additional inclusion criteria for BPD participants:

- valid F60.31 diagnosis according to ICD-10
- no F84 diagnosis according to ICD-10
- IQ estimate above 70

Additional inclusion criteria for comparison participants:

- IQ estimate above 70
- no psychiatric diagnoses
- no current psychotropic medication

Furthermore, we applied the following exclusion criteria:

- underage people or people older than 60 years
- inability to speak
- current or previous neurological disorder
- acute suicidality, self-harming or aggressive behaviour
- lack of legally binding informed consent

Testing was discontinued if participants withdrew their consent, exclusion criteria applied or in the case of technical difficulties.

S4 Procedures

The hobbies prompt was chosen to capture monologuing by ASD adults when talking about their interests⁵, while the mealplanning conversation required increased turn-taking and collaboration from both participants⁶. The order of the conversations was counterbalanced across participants and the experimenter left the room during the conversations. After both conversations were finished, they filled out a custom rapport questionnaire.

All conversations took place in stable artificial light to control for any biases caused by lighting changes⁷. Since some of the data was collected during the COVID-19 pandemic, we modified the setup to maximise the safety of our participants, for instance by placing a plexiglass between them. Furthermore, participants filled out several questionnaires to capture subjective

Table S1: Group comparisons

measurement	ASD	BPD	COMP	BPD vs. ASD	COMP vs. ASD	COMP vs. BPD
ADC_total	49.06 (± 17.17), n = 17	76.95 (± 14.98), n = 21	34.99 (± 23.86), n = 82	0.006*	0.243	0.000*
AQ_total	34.29 (± 6.18), n = 17	23.67 (± 4.72), n = 21	14.57 (± 5.27), n = 82	0.000*	0.000*	0.000*
BDI_total	15.94 (± 11.57), n = 17	24.81 (± 10.48), n = 21	3.99 (± 3.79), n = 82	0.715	0.000*	0.000*
BERT.acc	0.80 (± 0.07), n = 17	0.82 (± 0.09), n = 21	0.83 (± 0.07), n = 82	1.000	0.919	1.000
BERT.rt	5.79 (± 2.78), n = 17	3.84 (± 1.24), n = 21	3.33 (± 1.33), n = 82	0.453	0.000*	0.637
BSL_total	NaN ($\pm$ NA), n = 0	46.71 (± 21.44), n = 21	7.19 (± 6.84), n = 37	NA	NA	0.000*
IQ_estimate	116.47 (± 13.90), n = 17	109.60 (± 9.54), n = 20	112.77 (± 13.26), n = 82	1.000	1.000	1.000
SMS_total	5.47 (± 2.81), n = 17	11.67 (± 2.11), n = 21	9.80 (± 2.62), n = 82	0.000*	0.000*	0.027*
SPF_total	37.00 (± 7.20), n = 17	40.95 (± 8.36), n = 21	44.73 (± 5.82), n = 82	1.000	0.000*	0.328
TAS_total	61.76 (± 11.03), n = 17	55.33 (± 11.60), n = 21	37.55 (± 8.53), n = 82	1.000	0.000*	0.000*
age	37.59 (± 13.19), n = 17	28.38 (± 10.02), n = 21	28.89 (± 10.18), n = 82	0.506	0.243	1.000
plexi	0.82 (± 0.53), n = 17	0.85 (± 0.90), n = 13	0.86 (± 0.60), n = 43	1.000	1.000	1.000
rapport	12.24 (± 2.19), n = 17	11.95 (± 2.64), n = 21	12.33 (± 2.39), n = 81	1.000	1.000	1.000
video	0.71 (± 0.69), n = 17	0.81 (± 0.75), n = 21	0.63 (± 0.68), n = 82	1.000	1.000	1.000

clinical self-ratings, completed two intelligence assessments which were combined for the IQ estimate and performed the Berlin Emotion Recognition Test (BERT)⁸.

Audio and video streams were time-locked by aligning them based on the sound and visual of a movie clap denoting the start of the task. Videos and audio tracks were cut to 10min length using DaVinci Resolve.

S5 Measures

We collected the following self-report questionnaires:

- Autism-like traits: Autism Quotient, AQ⁹
- Empathy: Saarbrücker Persönlichkeitsfragebogen, SPF¹⁰, which is the German version of the interpersonal reactivity index
- Alexithymia: Toronto Alexithymia Scale, TAS²⁰¹¹
- Depressiveness: Beck Depression Inventory, BDI¹²
- Self-monitoring: Self-monitoring Scale, SMS¹³
- Movement difficulties: Adult Dyspraxia Checklist, ADC¹⁴

For all participants recruited for this project, we also collected the short version of the Borderline Symptom List, BSL-23¹⁵.

For 37 of the dyads, we used a plexiglass screen placed between the interaction partners on the table to decrease the chances of spreading an undetected infection. We applied a transparent anti-reflection foil to reduce any mirroring effects. During the conversation, participants took off their face masks. We asked these participants how much the plexiglass influenced them during the interactions (plexi; scale from 0 to 3). All participants were asked how much the cameras influenced their behaviour (video; scale from 0 to 3). Rapport is a sum of the following ratings, all on scales from 0 to 3, thus ranging from 0 to 15:

- How likeable was your interaction partner?
- How friendly was your interaction partner?
- How comfortable did you feel during the conversation?
- How smooth was the communication between you and your conversation partner?
- How well did your conversation partner respond to you?

S5.1 Group comparisons based on diagnostic status

Participants did not differ in age (ASD: $\bar{x} = 37.59 \pm 13.19$; BPD: $\bar{x} = 28.38 \pm 10.02$; COMP: $\bar{x} = 28.89 \pm 10.18$; BPD vs. ASD: pFDR = 0.506; ASD vs. COMP: pFDR = 0.243; BPD vs. COMP: pFDR = 1), IQ estimate (ASD: $\bar{x} = 116.47 \pm 13.90$; BPD: $\bar{x} = 109.60 \pm 9.54$; COMP: $\bar{x} = 112.77 \pm 13.26$; each pFDR = 1), but there was a trend for a difference in gender distribution (ASD: 6 female, 11 male; BPD: 14 female, 7 male; COMP: 52 female, 30 male; p = 0.078). However, our exploration of the influence of gender on decision scores in our support vector machines revealed no difference. Testing duration was two to three and a half hours.

ilabel

```
## gender ASD BPD COMP
##      fem   6  14   52
##      mal  11   7   30

##
## Pearson's Chi-squared test
##
## data:  tb.gen
## X-squared = 5.1108, df = 2, p-value = 0.07766
```

S6 Features

S6.1 Facial expressions extracted from OpenFace

We only included data of participants with a mean confidence of tracked frames greater than 75% and more than 90% successfully tracked frames. Facial expressions were captured as action units. We did not extract emotional expressions from these facial expressions as coherence between facial expressions and emotions is not a given and might be even less so for autistic people¹⁶.

For the calculation of synchronisation, we included rotational parameters (yaw, roll, pitch) as well as the same action units as in our previous study¹⁷:

- Mealplanning: 1, 2, 6, 7, 9, 14, 15, 17, 20, 25, 26 and 45
- Hobbies: 1, 2, 6, 7, 9, 15, 17, 20, 23, 25, 26 and 45

We also extracted total facial expressiveness as mean intensity of all action units for each interaction partner to be included in the *MovEx* and the *CROSSturn* models. For the *CROSSturn* model, we also included other action units, as listed below.

These correspond to the following movements:

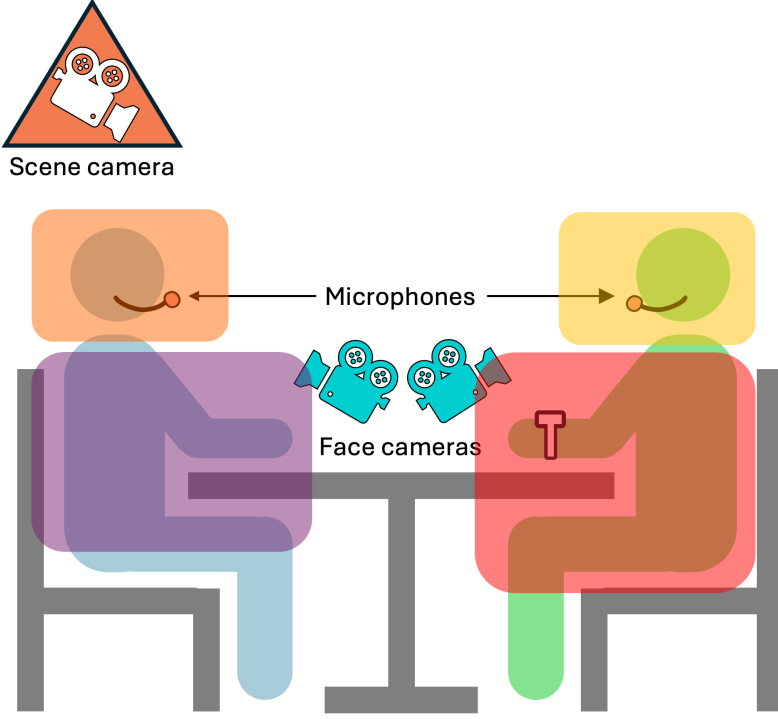
- AU1: inner brow raiser
- AU2: outer brow raiser
- AU4: brow lowerer (only *CROSSturn*)
- AU5: upper lid raiser (only *CROSSturn*)
- AU6: cheek raiser
- AU7: lid tightener
- AU9: nose wrinkler
- AU10: upper lip raiser (only *CROSSturn*)
- AU12: lip corner puller (only *CROSSturn*)
- AU14: dimpler
- AU15: lip corner depressor
- AU17: chin raiser
- AU20: lip stretcher
- AU23: lip tightener
- AU25: lips part
- AU26: jaw drop
- AU45: blink

Furthermore, we used translational head position parameters to infer head motion using the following formula with Δ_t referring to the respective frame-to-frame changes:

$$\text{head movement} = \sqrt{\Delta_t x^2 + \Delta_t y^2 + \Delta_t z^2}$$

S6.2 Motion quantity extracted using Motion Energy Analysis

This figure shows body (red and purple) and head (yellow and orange) regions of interests of each interaction partner separately:



There was always a space between the head and body region. Regions were chosen such that they cover the full range of motion throughout one conversation of one interaction partner. Thus, their sizes differed which is why we scaled all values.

In addition to using the motion quantity to compute synchronisation, we also extracted total movement in each region of interest for each interaction partner to be included in the *MovEx* model.

S6.3 Speech and turn-taking features

We extracted pitch using praat’s autocorrelation method, a technique widely recognized for its reliability and accuracy¹⁸. We implemented a two-step pitch extraction method, as outlined by Hirst¹⁹. First, to capture a broad range of frequencies, we set a low pitch floor of 50 Hz and a high pitch ceiling of 700 Hz, with a time step of 15 ms. All other parameters were set to Praat’s default values. Second, using these initial pitch values, we determined the first and third quartiles of pitch for each participant and task. We then used these quartiles to compute individual pitch floors and ceilings with the following algorithm:

$$\text{floor} = \min(0.75 \cdot Q_{1,hobbies}, 0.75 \cdot Q_{1,mealplanning})$$

$$\text{ceiling} = \max(2.5 \cdot Q_{3,hobbies}, 2.5 \cdot Q_{3,mealplanning})$$

We then used these individual pitch floors (range = 46 to 168Hz, mean = 109.7 ± 31.8) and ceilings (range = 250 to 839Hz, mean = 481.7 ± 147.4) to extract pitch. To ensure an equal number of frames for all participants, we maintained a consistent time step across all analyses. By default, praat calculates this time step using the following formula:

$$\text{timestep} = \frac{0.75}{\text{floor}}$$

Here, we used the same time step of 0.016 as in our previous study²⁰ which was determined based on the minimum individual pitch floor of that sample. Since the new sample did not include anyone with a lower pitch floor, the time step fits both samples.

Intensity was extracted by convolving the squared sound with a Gaussian analysis window. We used praat’s default values of minimum pitch 100Hz and time step of 0.01s.

To estimate synchrony, we extracted continuous pitch and intensity time series for every millisecond of the recording. For pitch extraction, we used consistent parameters across all participants instead of individualized settings. This was necessary because the analysis width depends on the pitch floor. Given the heterogeneity of our sample, we opted for a wide range

of considered frequencies, setting the pitch floor at 50Hz and the pitch ceiling at 700Hz. For intensity, we relied on Praat’s default values.

In the case of turn-based synchronization, we correlated the median pitch or intensity of each turn with the median pitch or intensity of the preceding turn.

Next, we used the uhm-o-meter^{21,22} to differentiate between periods of speaking and silence, identify syllables and extract several prosodic features (total number of syllables, total number of silent phases, duration of speaking as phonation time, speech rate as number of syllables per second, articulation rate as number of syllables per phonation time, average syllable duration and silence-to-turn ratio). The resulting speaking and silent instances were visually and aurally inspected to verify the accuracy of the algorithm.

S6.4 Cross-modal features

We captured two types of cross-modal features:

- Interpersonal synchronisation of one person’s head with the other person’s body movement and vice versa
- AU activation, body and head movement during listening and speaking

S6.5 Synchrony computations

We used the following settings for our windowed lagged cross-correlation (WLCC): For lag, step and window size, we stayed in line with previous literature²⁶. We applied Fisher’s Z-transformation and converted all values to their absolute²⁷. Furthermore, we computed turn-based adaptation of pitch and intensity.

Table S2: WLCC settings in seconds

measure	window	step	lag
Facial action units synchronisation	7	4	2
Body MEA synchronisation	30	15	5
Head MEA synchronisation	30	15	5
Intrapersonal synchrony	30	15	5
Pitch synchrony	16	8	2
Intensity synchrony	16	8	2

For each window, the maximum correlation value was chosen out of all relevant lags (peack-picking). We cross-correlated head movements (from OpenFace) with body motion energy time series (from MEA) to estimate intrapersonal synchrony.

S6.6 Feature lists

This list shows all features without the information of conversation, i.e., each of these features was added twice to the model, once from the mealplanning and once from the hobbies conversation. The number of features per model is displayed as well. For many of the extracted features, we calculated summary scores some which are indicated by abbreviations (mean, md = median, sd = standard deviation, min = minimum, max = maximum, ske = skewness and kurtosis)

Table S3: List of all mealplanning features of the base models

model	features
BODYsync (14 features)	min_M_bodysync, max_M_bodysync, sd_M_bodysync, mean_M_bodysync, md_M_bodysync, skew_M_bodysync, kurtosis_M_bodysync
CROSSsync (28 features)	min_M_LOF, min_M_ROF, max_M_LOF, max_M_ROF, md_M_LOF, md_M_ROF, mean_M_LOF, mean_M_ROF, sd_M_LOF, sd_M_ROF, kurtosis_M_LOF, kurtosis_M_ROF, skew_M_LOF, skew_M_ROF

CROSSturn
(532 features)

self_min_M_AU01_r, self_min_M_AU02_r, self_min_M_AU04_r, self_min_M_AU05_r,
self_min_M_AU06_r, self_min_M_AU07_r, self_min_M_AU09_r, self_min_M_AU10_r,
self_min_M_AU12_r, self_min_M_AU14_r, self_min_M_AU15_r, self_min_M_AU17_r,
self_min_M_AU20_r, self_min_M_AU23_r, self_min_M_AU25_r, self_min_M_AU26_r,
self_min_M_AU45_r, self_min_M_MEA_body, self_min_M_MEA_head, self_max_M_AU01_r,
self_max_M_AU02_r, self_max_M_AU04_r, self_max_M_AU05_r, self_max_M_AU06_r,
self_max_M_AU07_r, self_max_M_AU09_r, self_max_M_AU10_r, self_max_M_AU12_r,
self_max_M_AU14_r, self_max_M_AU15_r, self_max_M_AU17_r, self_max_M_AU20_r,
self_max_M_AU23_r, self_max_M_AU25_r, self_max_M_AU26_r, self_max_M_AU45_r,
self_max_M_MEA_body, self_max_M_MEA_head, self_md_M_AU01_r, self_md_M_AU02_r,
self_md_M_AU04_r, self_md_M_AU05_r, self_md_M_AU06_r, self_md_M_AU07_r, self_md_M_AU09_r,
self_md_M_AU10_r, self_md_M_AU12_r, self_md_M_AU14_r, self_md_M_AU15_r, self_md_M_AU17_r,
self_md_M_AU20_r, self_md_M_AU23_r, self_md_M_AU25_r, self_md_M_AU26_r, self_md_M_AU45_r,
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self_mean_M_AU04_r, self_mean_M_AU05_r, self_mean_M_AU06_r, self_mean_M_AU07_r,
self_mean_M_AU09_r, self_mean_M_AU10_r, self_mean_M_AU12_r, self_mean_M_AU14_r,
self_mean_M_AU15_r, self_mean_M_AU17_r, self_mean_M_AU20_r, self_mean_M_AU23_r,
self_mean_M_AU25_r, self_mean_M_AU26_r, self_mean_M_AU45_r, self_mean_M_MEA_body,
self_mean_M_MEA_head, self_sd_M_AU01_r, self_sd_M_AU02_r, self_sd_M_AU04_r,
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self_sd_M_MEA_head, self_kurtosis_M_AU01_r, self_kurtosis_M_AU02_r, self_kurtosis_M_AU04_r,
self_kurtosis_M_AU05_r, self_kurtosis_M_AU06_r, self_kurtosis_M_AU07_r, self_kurtosis_M_AU09_r,
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self_skew_M_AU05_r, self_skew_M_AU06_r, self_skew_M_AU07_r, self_skew_M_AU09_r,
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other_md_M_AU04_r, other_md_M_AU05_r, other_md_M_AU06_r, other_md_M_AU07_r,
other_md_M_AU09_r, other_md_M_AU10_r, other_md_M_AU12_r, other_md_M_AU14_r,
other_md_M_AU15_r, other_md_M_AU17_r, other_md_M_AU20_r, other_md_M_AU23_r,
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other_mean_M_AU05_r, other_mean_M_AU06_r, other_mean_M_AU07_r, other_mean_M_AU09_r,
other_mean_M_AU10_r, other_mean_M_AU12_r, other_mean_M_AU14_r, other_mean_M_AU15_r,
other_mean_M_AU17_r, other_mean_M_AU20_r, other_mean_M_AU23_r, other_mean_M_AU25_r,
other_mean_M_AU26_r, other_mean_M_AU45_r, other_mean_M_MEA_body, other_mean_M_MEA_head,
other_sd_M_AU01_r, other_sd_M_AU02_r, other_sd_M_AU04_r, other_sd_M_AU05_r,
other_sd_M_AU06_r, other_sd_M_AU07_r, other_sd_M_AU09_r, other_sd_M_AU10_r,
other_sd_M_AU12_r, other_sd_M_AU14_r, other_sd_M_AU15_r, other_sd_M_AU17_r,
other_sd_M_AU20_r, other_sd_M_AU23_r, other_sd_M_AU25_r, other_sd_M_AU26_r,
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other_skew_M_AU12_r, other_skew_M_AU14_r, other_skew_M_AU15_r, other_skew_M_AU17_r,
other_skew_M_AU20_r, other_skew_M_AU23_r, other_skew_M_AU25_r, other_skew_M_AU26_r,
other_skew_M_AU45_r, other_skew_M_MEA_body, other_skew_M_MEA_head

FACEsync (168 features)	min_M_AU01_r, max_M_AU01_r, sd_M_AU01_r, mean_M_AU01_r, md_M_AU01_r, skew_M_AU01_r, kurtosis_M_AU01_r, min_M_AU02_r, max_M_AU02_r, sd_M_AU02_r, mean_M_AU02_r, md_M_AU02_r, skew_M_AU02_r, kurtosis_M_AU02_r, min_M_AU06_r, max_M_AU06_r, sd_M_AU06_r, mean_M_AU06_r, md_M_AU06_r, skew_M_AU06_r, kurtosis_M_AU06_r, min_M_AU07_r, max_M_AU07_r, sd_M_AU07_r, mean_M_AU07_r, md_M_AU07_r, skew_M_AU07_r, kurtosis_M_AU07_r, min_M_AU09_r, max_M_AU09_r, sd_M_AU09_r, mean_M_AU09_r, md_M_AU09_r, skew_M_AU09_r, kurtosis_M_AU09_r, min_M_AU14_r, max_M_AU14_r, sd_M_AU14_r, mean_M_AU14_r, md_M_AU14_r, skew_M_AU14_r, kurtosis_M_AU14_r, min_M_AU15_r, max_M_AU15_r, sd_M_AU15_r, mean_M_AU15_r, md_M_AU15_r, skew_M_AU15_r, kurtosis_M_AU15_r, min_M_AU17_r, max_M_AU17_r, sd_M_AU17_r, mean_M_AU17_r, md_M_AU17_r, skew_M_AU17_r, kurtosis_M_AU17_r, min_M_AU20_r, max_M_AU20_r, sd_M_AU20_r, mean_M_AU20_r, md_M_AU20_r, skew_M_AU20_r, kurtosis_M_AU20_r, min_M_AU25_r, max_M_AU25_r, sd_M_AU25_r, mean_M_AU25_r, md_M_AU25_r, skew_M_AU25_r, kurtosis_M_AU25_r, min_M_AU26_r, max_M_AU26_r, sd_M_AU26_r, mean_M_AU26_r, md_M_AU26_r, skew_M_AU26_r, kurtosis_M_AU26_r, min_M_AU45_r, max_M_AU45_r, sd_M_AU45_r, mean_M_AU45_r, md_M_AU45_r, skew_M_AU45_r, kurtosis_M_AU45_r
HEADsync (56 features)	min_M_headsync, max_M_headsync, sd_M_headsync, mean_M_headsync, md_M_headsync, skew_M_headsync, kurtosis_M_headsync, min_M_pose_Rxsync, max_M_pose_Rxsync, sd_M_pose_Rxsync, mean_M_pose_Rxsync, md_M_pose_Rxsync, skew_M_pose_Rxsync, kurtosis_M_pose_Rxsync, min_M_pose_Rysync, max_M_pose_Rysync, sd_M_pose_Rysync, mean_M_pose_Rysync, md_M_pose_Rysync, skew_M_pose_Rysync, kurtosis_M_pose_Rysync, min_M_pose_Rzsync, max_M_pose_Rzsync, sd_M_pose_Rzsync, mean_M_pose_Rzsync, md_M_pose_Rzsync, skew_M_pose_Rzsync, kurtosis_M_pose_Rzsync
INTRAsync (14 features)	min_M_intra, max_M_intra, sd_M_intra, mean_M_intra, md_M_intra, skew_M_intra, kurtosis_M_intra
MovEx (6 features)	M_body_total_movement, M_head_total_movement, mean_intensity_M
Speech (30 features)	dyad_pit_sync_MEA_M_speech, dyad_int_sync_MEA_M_speech, dyad_spr_M_speech, dyad_str_M_speech, dyad_ttg_M_speech, dyad_no_turns_M_speech, nsyll_M_speech, npause_M_speech, pho_M_speech, art_M_speech, pit_sync_M_speech, int_sync_M_speech, art_sync_M_speech, pit_var_M_speech, int_var_M_speech

S7 Support vector machines

The Speech model mirrors the model used for classification between interaction partners in ASD-involved and COMP interactions in Plank et al.²⁰. Similarly, five of our eight base models, namely BODYsync, FACEsync, HEADsync, INTRAsync and MovEx, as well as our stacking model mirror the models in Koehler et al.¹⁷. All models were performed with the same preprocessing and classification settings as in the previous studies unless specified here.

All models were preprocessed and analysed separately. Preprocessing for the base models included scaling the data from -1 to 1 and pruning non-informative features. Here, we slightly deviate from Koehler et al.¹⁷ who scaled from 0 to 1 to stay in line with Plank et al.²⁰ and the NeuroMiner recommendation. Additionally, for all base models with more than 50 features, we performed a principal component analysis, retaining the principal components explaining at least 80% of the variance, before applying pruning again. Furthermore, we used ensemble learning to choose the models performing in the top 50%, which deviates from Plank et al.²⁰ but is in line with Koehler et al.¹⁷.

We used a repeated, nested, stratified cross-validation structure where the inner loop allows for hyper-parameter tuning to be separate from the model evaluation in the outer loop. We implemented ten folds with ten permutations in the outer loop and nine folds with one permutation in the inner loop. Due to the dyadic nature of our data, we specifically constructed the cross-validation structure such that both interaction partners of one dyad are always in the same fold. Base models were trained using LIBLINEAR Support Vector L2-regularised L2-loss classification algorithms with balanced accuracy as the performance criterion and using hyperplane weighing for uneven group sizes. The features of the stacking model, which are the decision scores of the base models, were entered into an L1-loss LIBSVM algorithm with a Gaussian kernel and use single node learning.

S8 Model performance

S8.1 Distinguishing ASD-involved from comparison interactions

All base models except the *HEADsync* and the *INTRAsync* models performed significantly better than chance according to permutation tests (*HEADsync*: $p_{FDR} = 1$; *INTRAsync*: $p_{FDR} = 1$). For this comparison, the *FACEsync* model achieved the highest balanced accuracy, specifically 74.9% (79.4% sensitivity; 70.5% specificity), closely followed by the *CROSSsync* model with 71.3% balanced accuracy (67.6% sensitivity; 75% specificity). The stacking model had a significantly higher

balanced accuracy of 80.6% (79.4% sensitivity; 81.8% specificity). Here, seven ASD-involved interactions were misclassified as non-clinical, while zero non-clinical interactions were mistaken as ASD-involved interactions. The rest of the interaction partners, 63 in total, were classified accurately. The stacking did not perform significantly better than the best base model (*FACEsync*: $p_{FDR} = 0.112$) but outperformed all other base models.

S8.2 Distinguishing BPD-involved from COMP interactions

While developing an algorithm for technology-assisted diagnostics of BPD was not the explicit goal of this research project, we explored the application of our features to the classification between BPD-involved and COMP interactions. Despite the features being chosen with symptoms and characteristics of ASD in mind, the *CROSSsturn*, *FACEsync*, *HEADsync* and *Speech* models performed above chance in this comparison (*BODYsync*: $p_{FDR} = 1$; *CROSSsync*: $p_{FDR} = 0.282$; *INTRAsync*: $p_{FDR} = 1$; *MovEx*: $p_{FDR} = 0.242$). Specifically, the *HEADsync* model achieved 68.7% balanced accuracy (71.4; 65.9% specificity), the *FACEsync* 64% (64.3; 65.9% specificity), the *Speech* 61.6% (59.5; 63.6% specificity) and the *CROSSsturn* 55.7% (52.4; 59.1% specificity). The stacking model performed comparable to the *HEADsync* and the *MovEx* model but outperformed the other base models, reaching 65.3% balanced accuracy (73.8; 56.8% specificity). Thus, the stacking model only misclassified eleven BPD-involved interactions as COMP, but 19 COMP interactions were labelled as BPD-involved.

S8.3 One-versus-One comparisons

Table S4: Performance of all the models in One-versus-One comparisons

comparison	model	sens	spec	BAC	AUC	p.fdr	sig
ASD-COMP vs BPD-COMP	BODYsync	32.353	47.619	39.986	0.361	1.000	
ASD-COMP vs BPD-COMP	CROSSsync	52.941	69.048	60.994	0.678	0.136	
ASD-COMP vs BPD-COMP	CROSSsturn	67.647	85.714	76.681	0.749	0.000	*
ASD-COMP vs BPD-COMP	FACEsync	76.471	59.524	67.997	0.719	0.000	*
ASD-COMP vs BPD-COMP	HEADsync	55.882	57.143	56.513	0.609	0.936	
ASD-COMP vs BPD-COMP	INTRAsync	38.235	54.762	46.499	0.455	1.000	
ASD-COMP vs BPD-COMP	MovEx	58.824	78.571	68.698	0.758	0.027	*
ASD-COMP vs BPD-COMP	Speech	64.706	76.190	70.448	0.771	0.000	*
ASD-COMP vs BPD-COMP	STACK	70.588	92.857	81.723	0.805	NaN	NA
ASD-COMP vs COMP-COMP	BODYsync	58.824	61.364	60.094	0.607	0.000	*
ASD-COMP vs COMP-COMP	CROSSsync	67.647	75.000	71.324	0.763	0.000	*
ASD-COMP vs COMP-COMP	CROSSsturn	44.118	72.727	58.422	0.627	0.000	*
ASD-COMP vs COMP-COMP	FACEsync	79.412	70.454	74.933	0.780	0.000	*
ASD-COMP vs COMP-COMP	HEADsync	58.824	68.182	63.503	0.608	1.000	
ASD-COMP vs COMP-COMP	INTRAsync	26.471	40.909	33.690	0.297	1.000	
ASD-COMP vs COMP-COMP	MovEx	64.706	72.727	68.717	0.777	0.000	*
ASD-COMP vs COMP-COMP	Speech	64.706	72.727	68.717	0.673	0.000	*
ASD-COMP vs COMP-COMP	STACK	79.412	81.818	80.615	0.845	NaN	NA
BPD-COMP vs COMP-COMP	BODYsync	28.571	36.364	32.468	0.308	1.000	
BPD-COMP vs COMP-COMP	CROSSsync	50.000	61.364	55.682	0.594	0.282	
BPD-COMP vs COMP-COMP	CROSSsturn	52.381	59.091	55.736	0.569	0.045	*
BPD-COMP vs COMP-COMP	FACEsync	64.286	63.636	63.961	0.670	0.007	*
BPD-COMP vs COMP-COMP	HEADsync	71.429	65.909	68.669	0.736	0.000	*
BPD-COMP vs COMP-COMP	INTRAsync	57.143	54.545	55.844	0.516	1.000	
BPD-COMP vs COMP-COMP	MovEx	69.048	70.454	69.751	0.686	0.242	
BPD-COMP vs COMP-COMP	Speech	59.524	63.636	61.580	0.693	0.007	*
BPD-COMP vs COMP-COMP	STACK	73.810	56.818	65.314	0.760	NaN	NA

S8.4 Multi-group comparisons

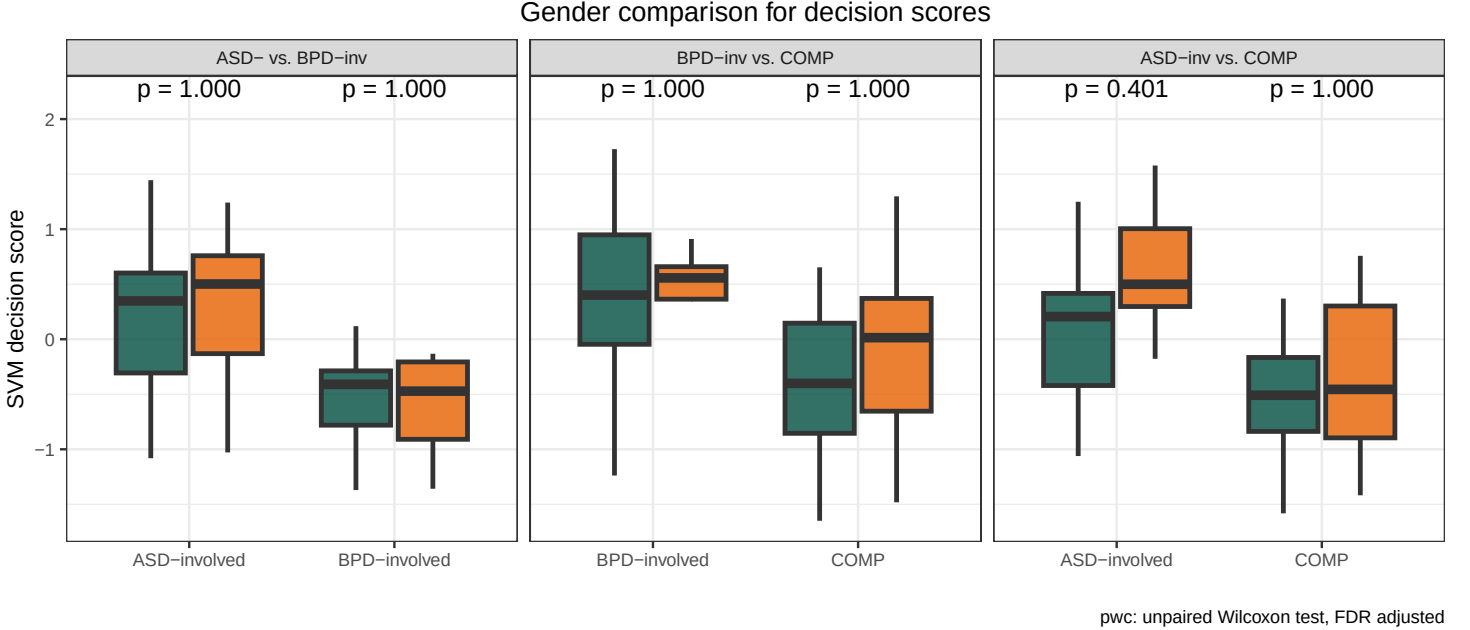
Table S5: Performance of all the models in Multi-group comparisons

comparison	model	BAC	p.fdr	sig
MultiGroup	BODYsync	48.6	1.000	
MultiGroup	CROSSsync	58.1	0.000	*
MultiGroup	CROSSsturn	57.3	0.000	*
MultiGroup	FACEsync	57.6	0.007	*
MultiGroup	HEADsync	54.6	0.000	*

MultiGroup	INTRAsync	49.5	1.000	
MultiGroup	MovEx	60.8	0.000	*
MultiGroup	Speech	63.9	0.000	*
MultiGroup	STACK	63.2	NaN	NA

S8.5 Gender comparisons

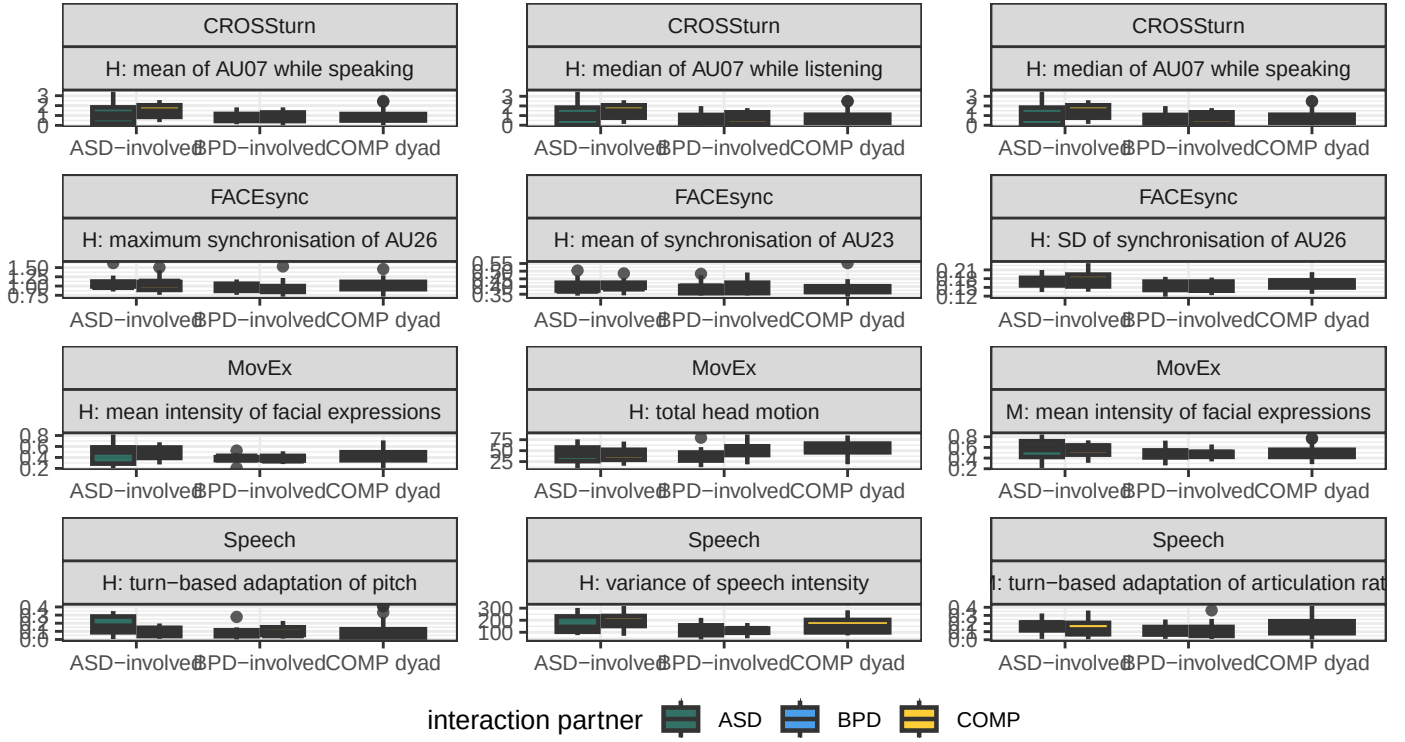
Did the models perform better for one gender than the other? We perform unpaired Wilcoxon tests for the labels and models separately.



S9 Exploring influential features

To explore the features in the interaction partners themselves, we plotted the three most influential features for the distinction between ASD-involved and BPD-involved as well as for the distinction between ASD-involved and non-clinical. We focused on the *CROSSturn*, *FACEsync*, *MovEx* and *Speech* models and determined the most influential features based on their feature weights. For the *Speech* model, we focused on individual features. Visual exploration of these features revealed COMP interaction partners in the ASD- or BPD-involved dyads adapting their behaviour to their clinical counterpart in some of the features. For instance, group comparisons revealed that COMP interaction partners did not differ from their clinical interaction partners in the variance of their speech intensity in the hobbies conversation (ASD-involved: $p_{HSD} = 0.711$; BPD-involved: $p_{HSD} = 0.999$), but COMP interaction partners' behaviour differed between COMP interaction partners in BPD-involved and COMP interaction partners in ASD-involved interactions ($p_{HSD} < 0.05$). Furthermore, COMP interaction partners in non-clinical interactions behaved comparably to COMP interaction partners in both clinical interactions (ASD-involved: $p_{HSD} = 0.2$; BPD-involved: $p_{HSD} = 0.203$). A similar pattern can be observed for the standard deviation of synchronisation of AU26, which measures jaw dropping, in the hobbies conversation with COMP interaction partners in clinical interactions not differing from their interaction partners (ASD-involved: $p_{HSD} = 0.999$; BPD-involved: $p_{HSD} = 0.936$), but COMP interaction partners in BPD-involved interactions differing from COMP interaction partners in ASD-involved interactions ($p_{HSD} < 0.05$). COMP interaction partners also showed comparable total head motion in the hobbies conversation to their interaction partners (ASD-involved: $p_{HSD} = 1$; BPD-involved: $p_{HSD} = 0.151$), but here they also did not differ between clinical interactions ($p_{HSD} = 0.57$). However, COMP interaction partners in non-clinical interactions exhibited more head motion than COMP interaction partners in ASD-involved interactions ($p_{HSD} = 0.04$), but not from COMP interaction partners in BPD-involved interactions ($p_{HSD} = 0.726$).

In the following boxplot, we focus here on the most influential features from base models with the highest balanced accuracy when classifying between interaction partners from BPD- or ASD-involved interactions. Comparison adults without any psychiatric diagnoses (COMP) seem to adjust their behaviour to their BPD or ASD interaction partner for some features, for instance the variance of their speech intensity. AU = facial action unit (AU07 = lid tightener; AU23 = lip tightener; AU26 = jaw drop); H = hobbies conversation; M = mealplanning conversation; SD = standard deviation.

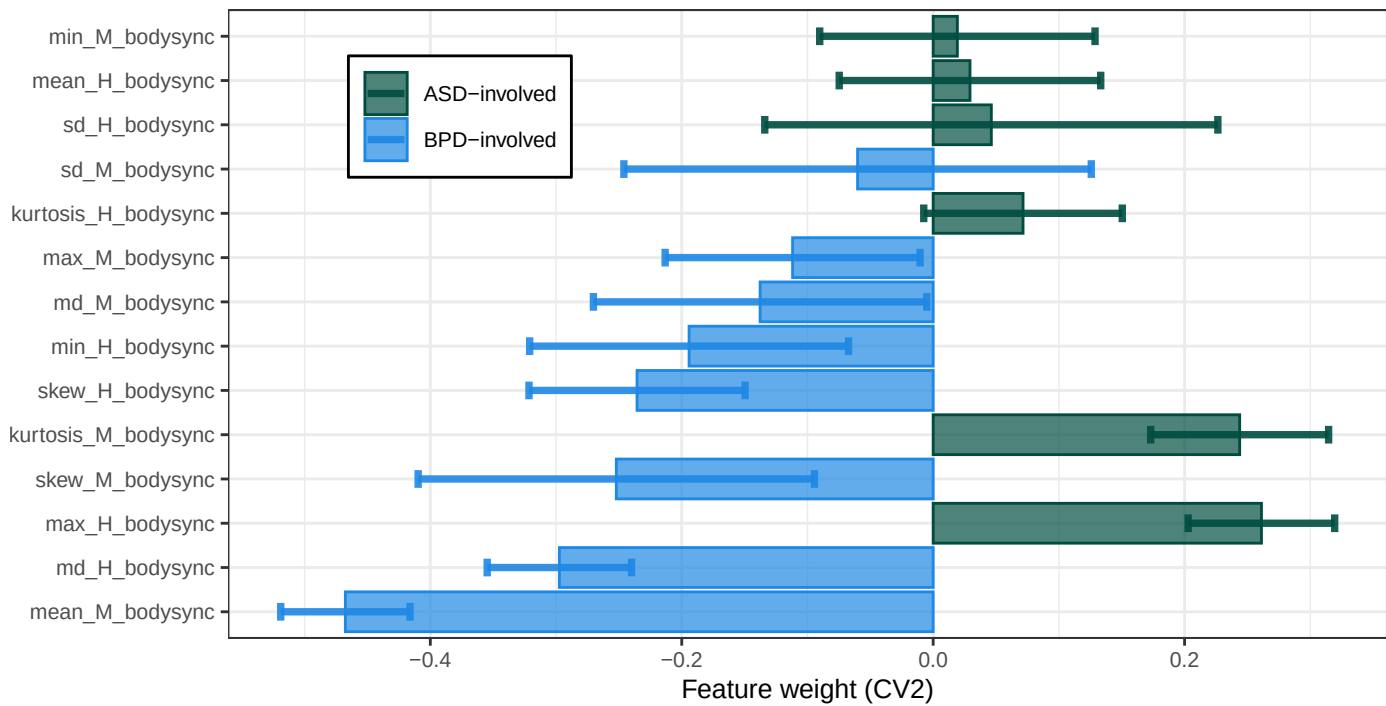


S10 Model visualisations

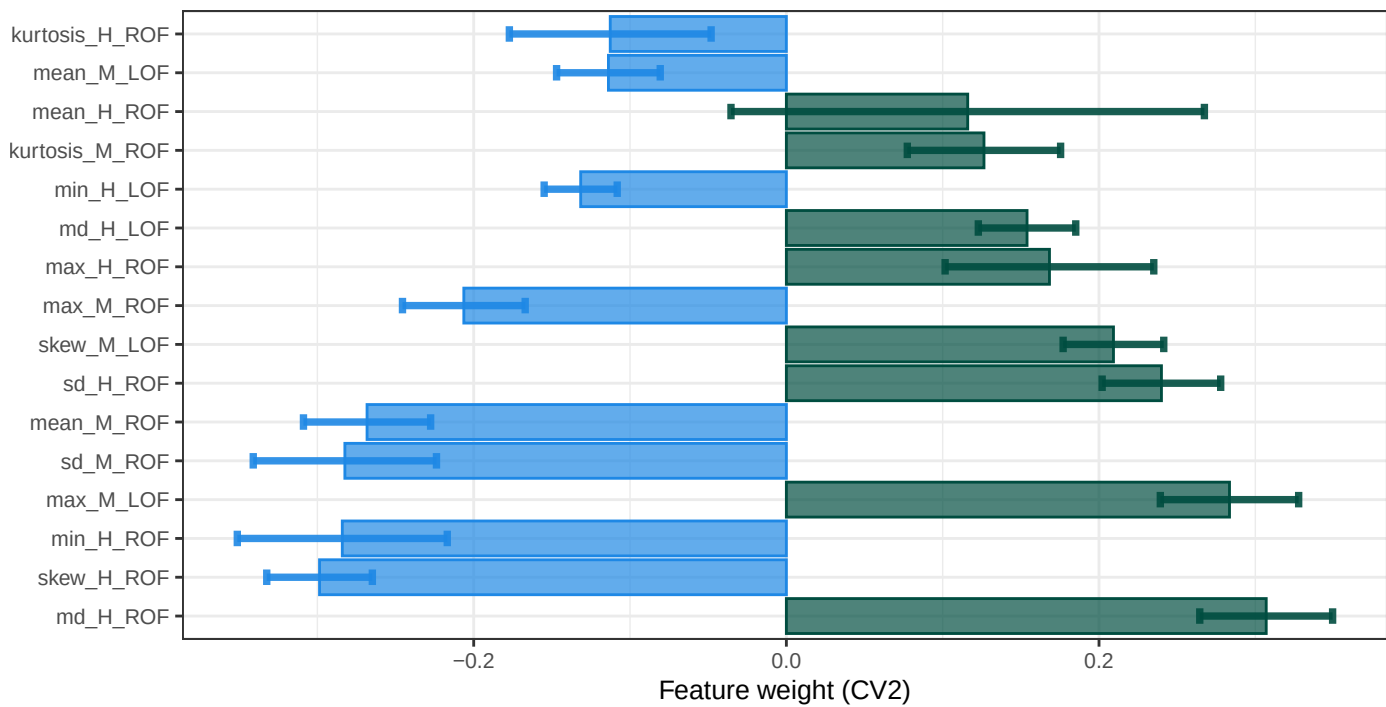
The following figures were created inspired by NeuroMiner⁴ visualisations and show the sign-based consistency²⁸ as well as the feature weights of the models distinguishing between interaction partners from ASD- and BPD-involved interactions. For models with a large number of features, the top 16 features are plotted.

S10.1 ASD-involved versus BPD-involved classifiers

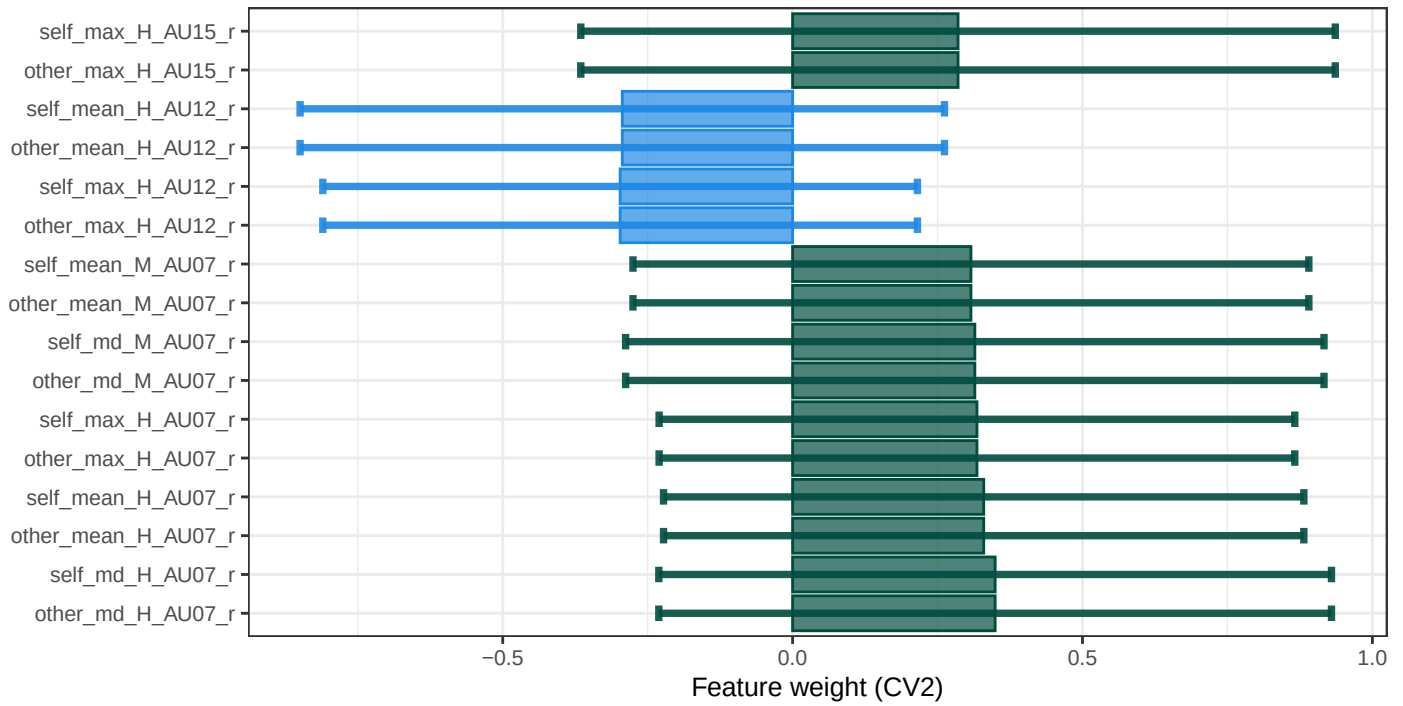
BODYsync



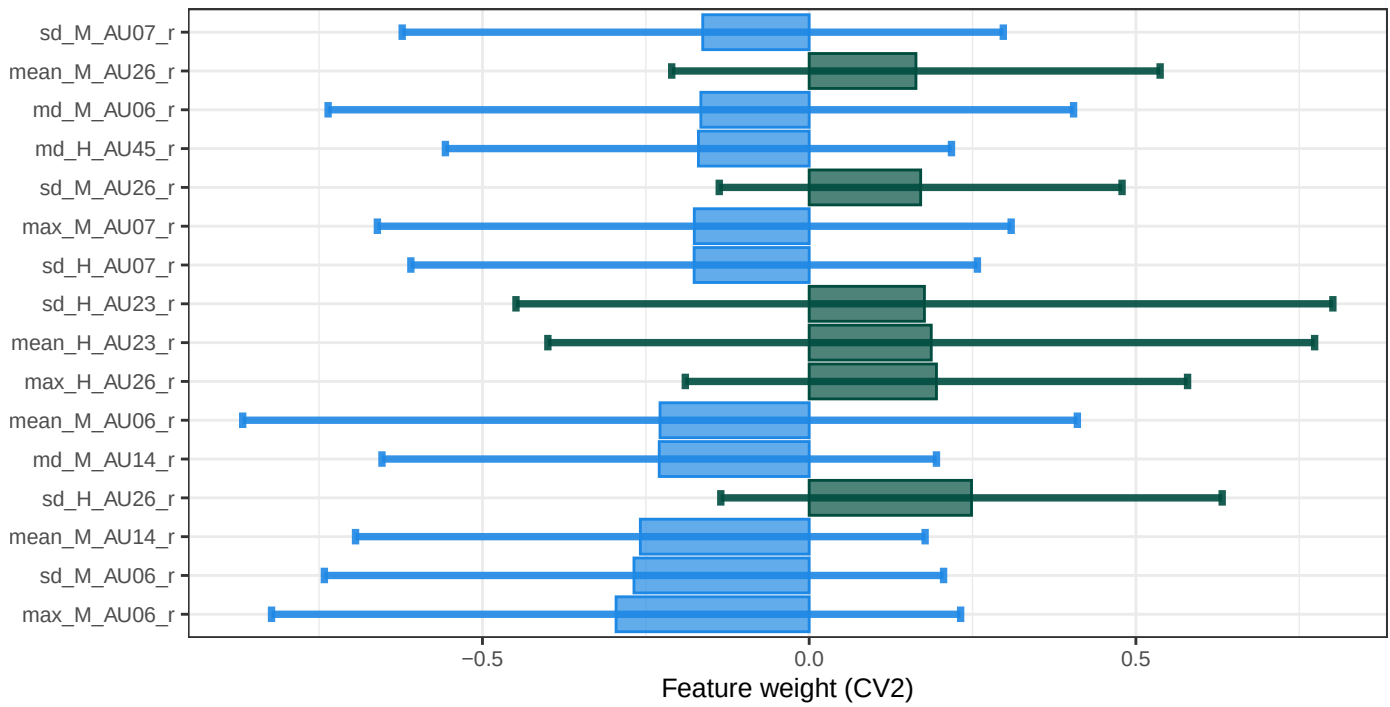
CROSSsync



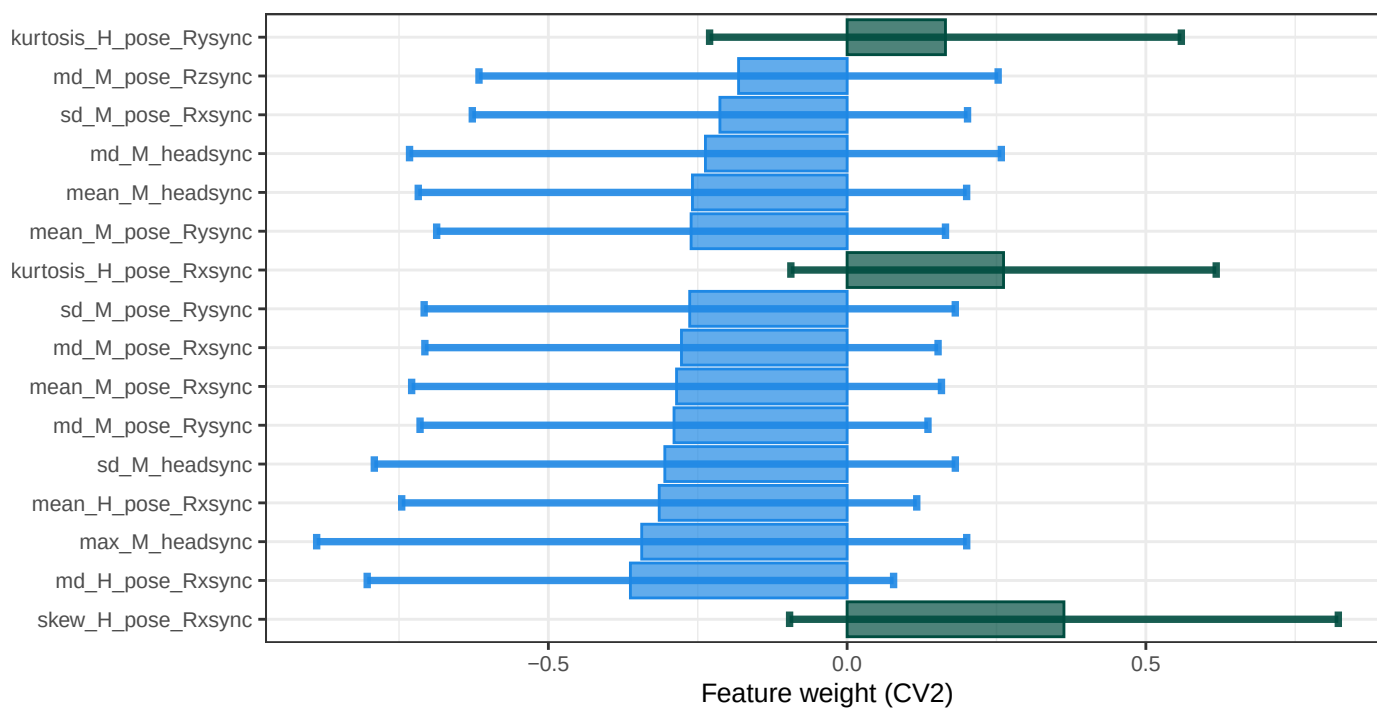
CROSSturn



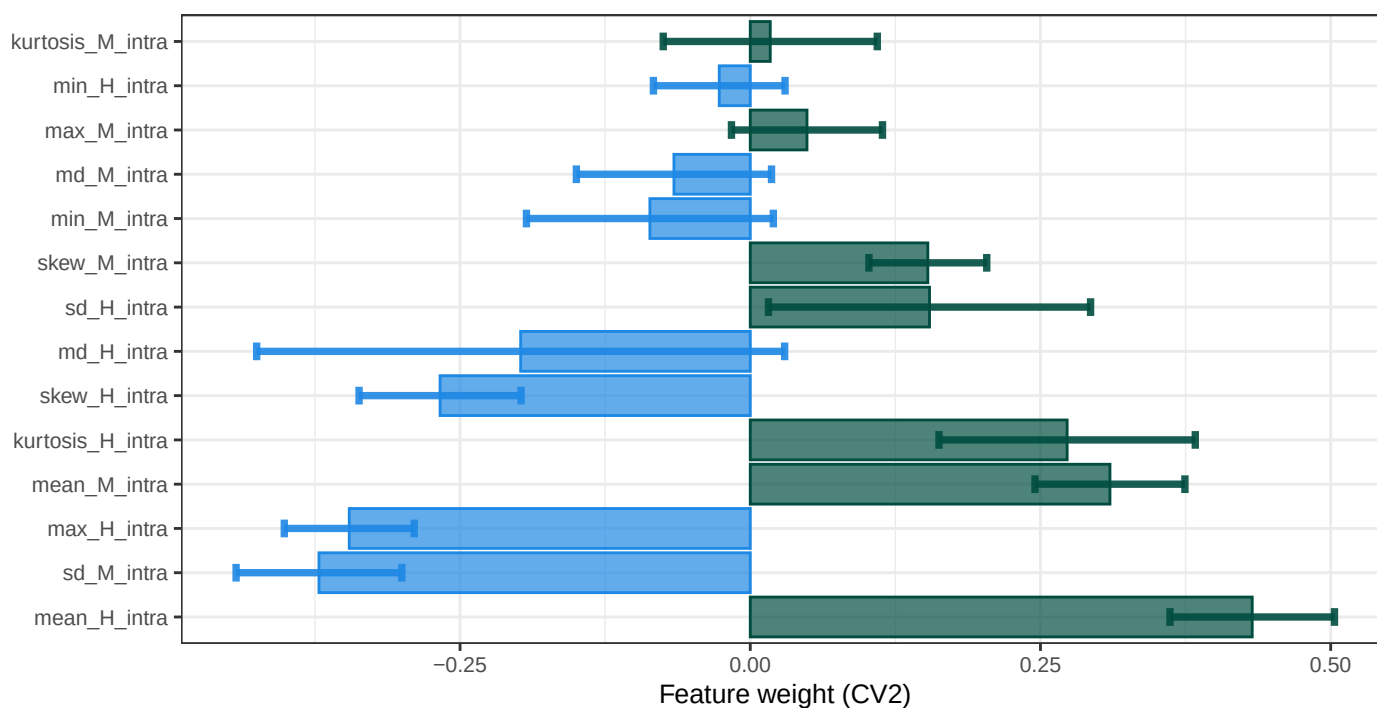
FACEsync



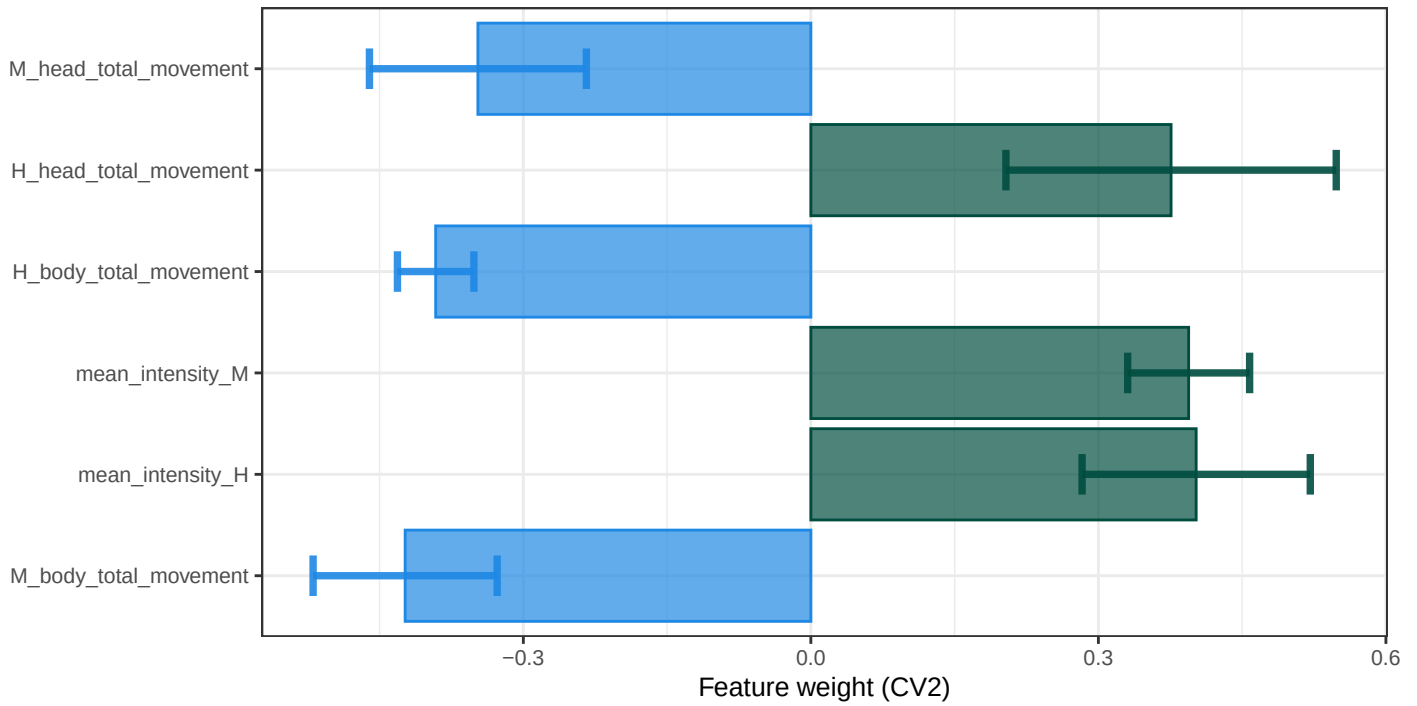
HEADsync



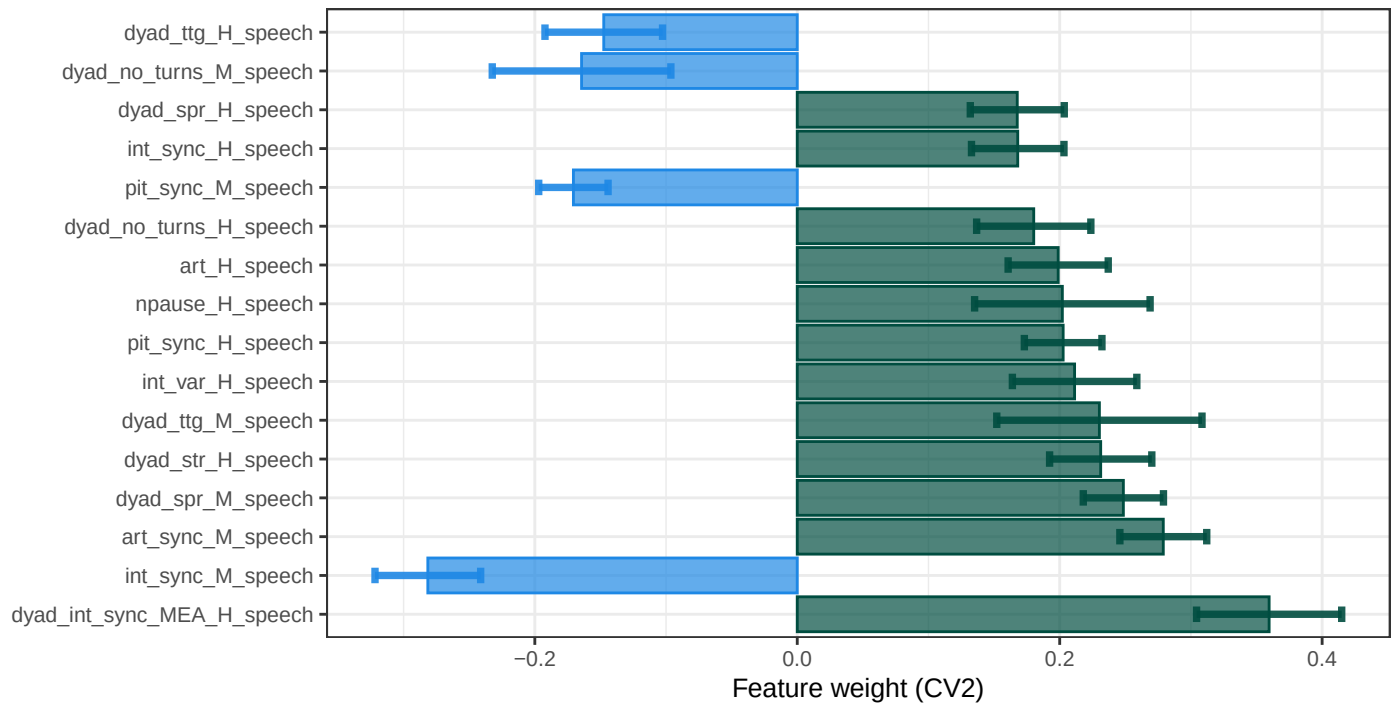
INTRAsync



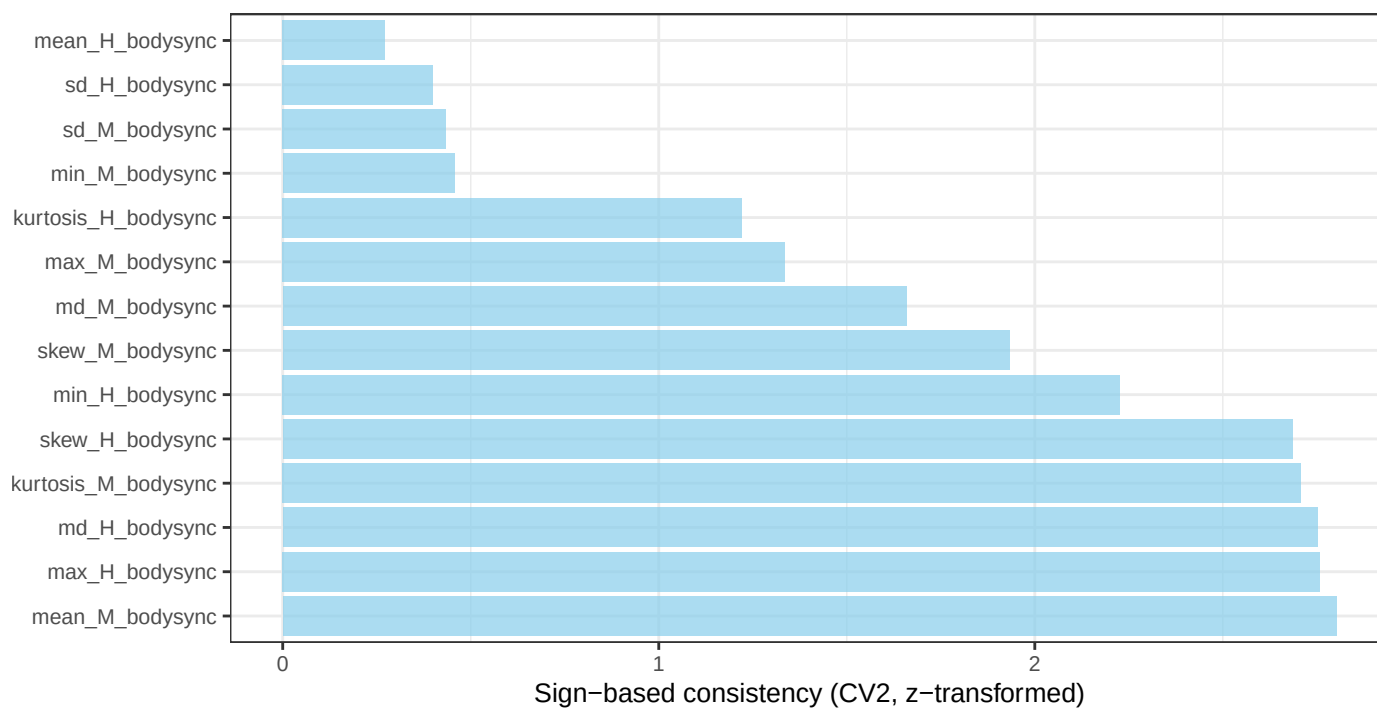
MovEx



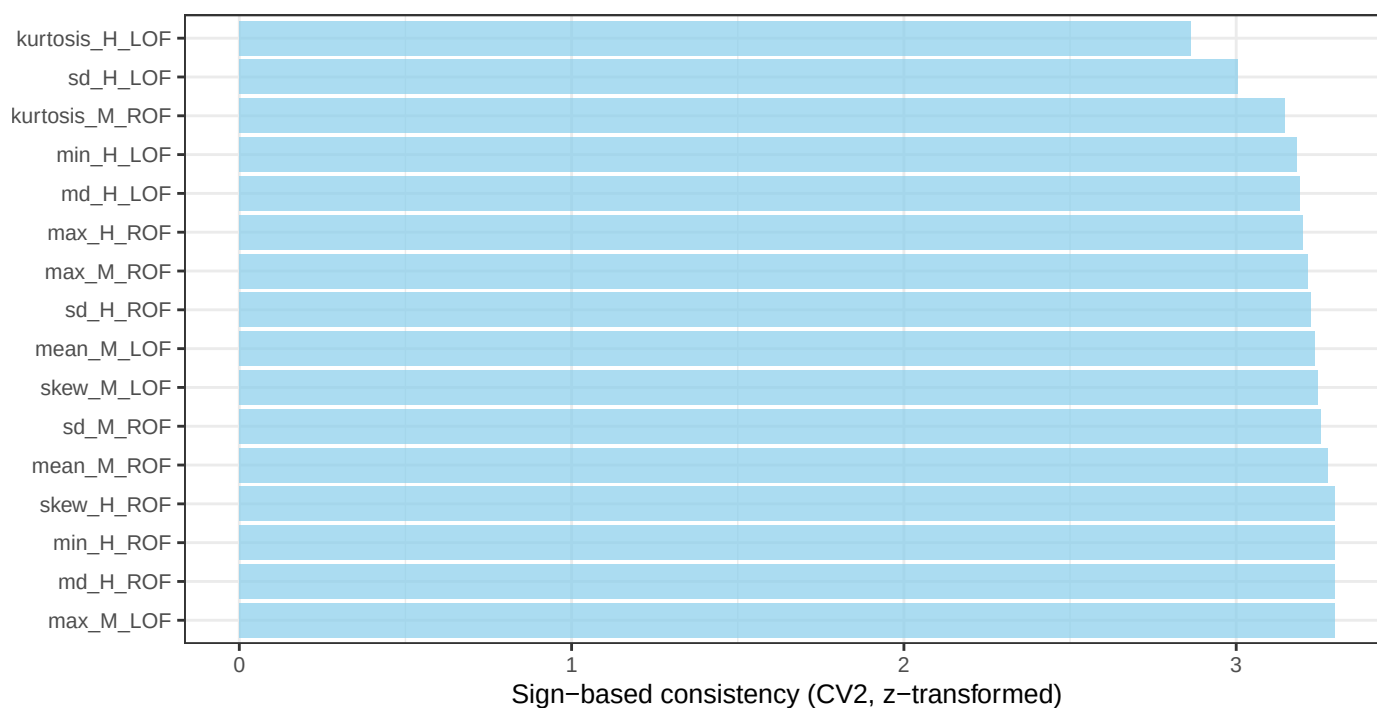
Speech



BODYsync



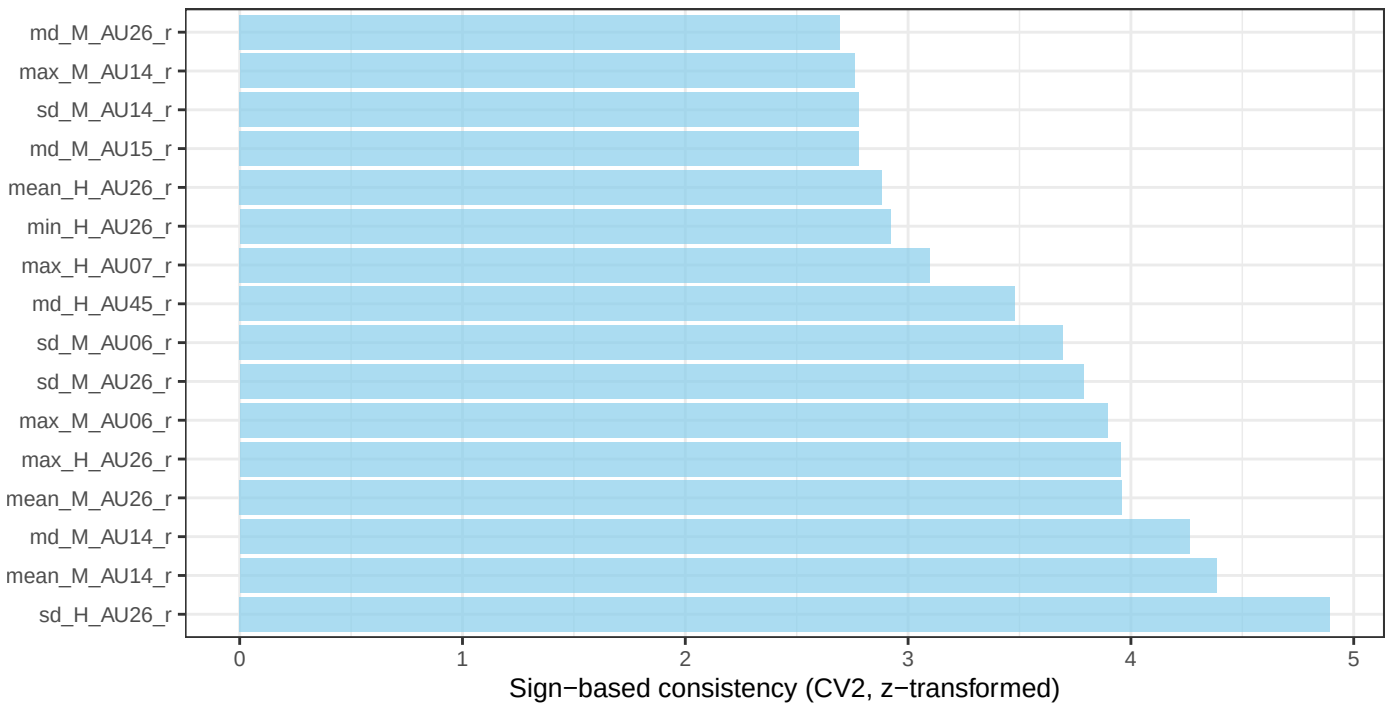
CROSSsync



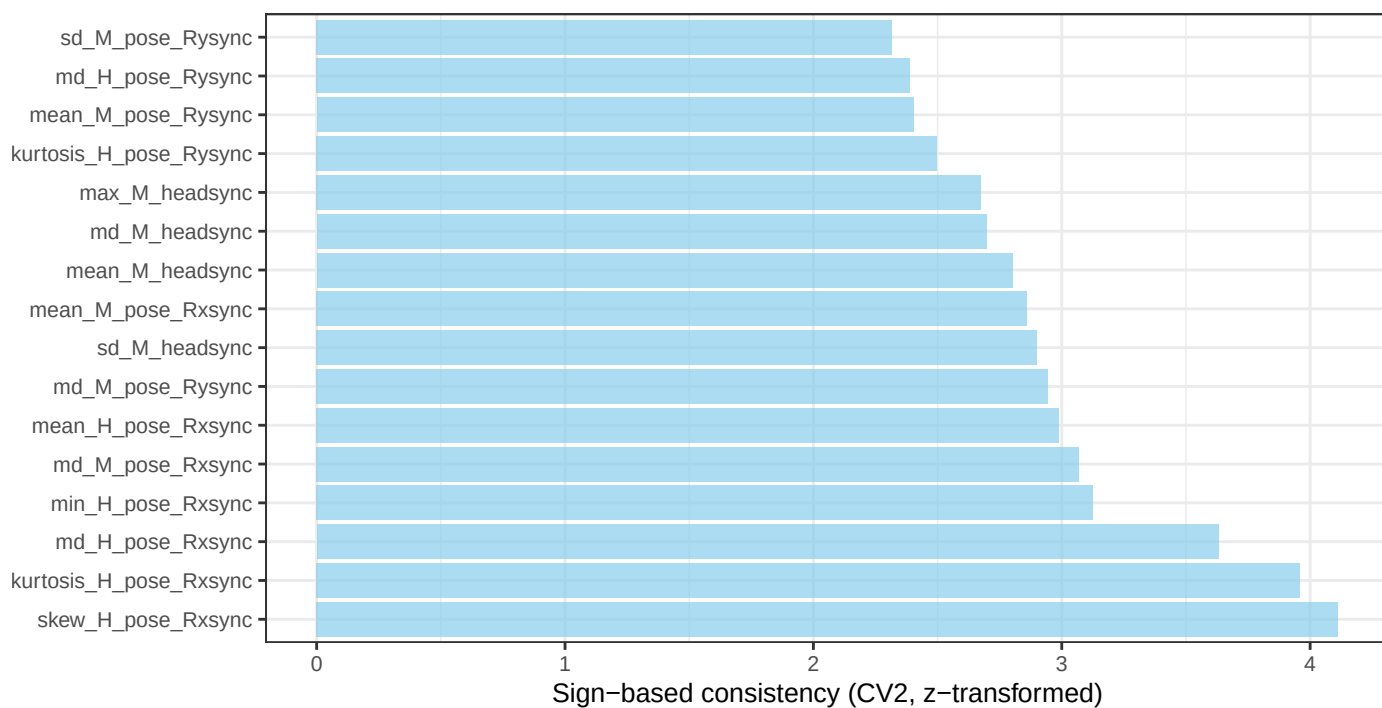
CROSSturn



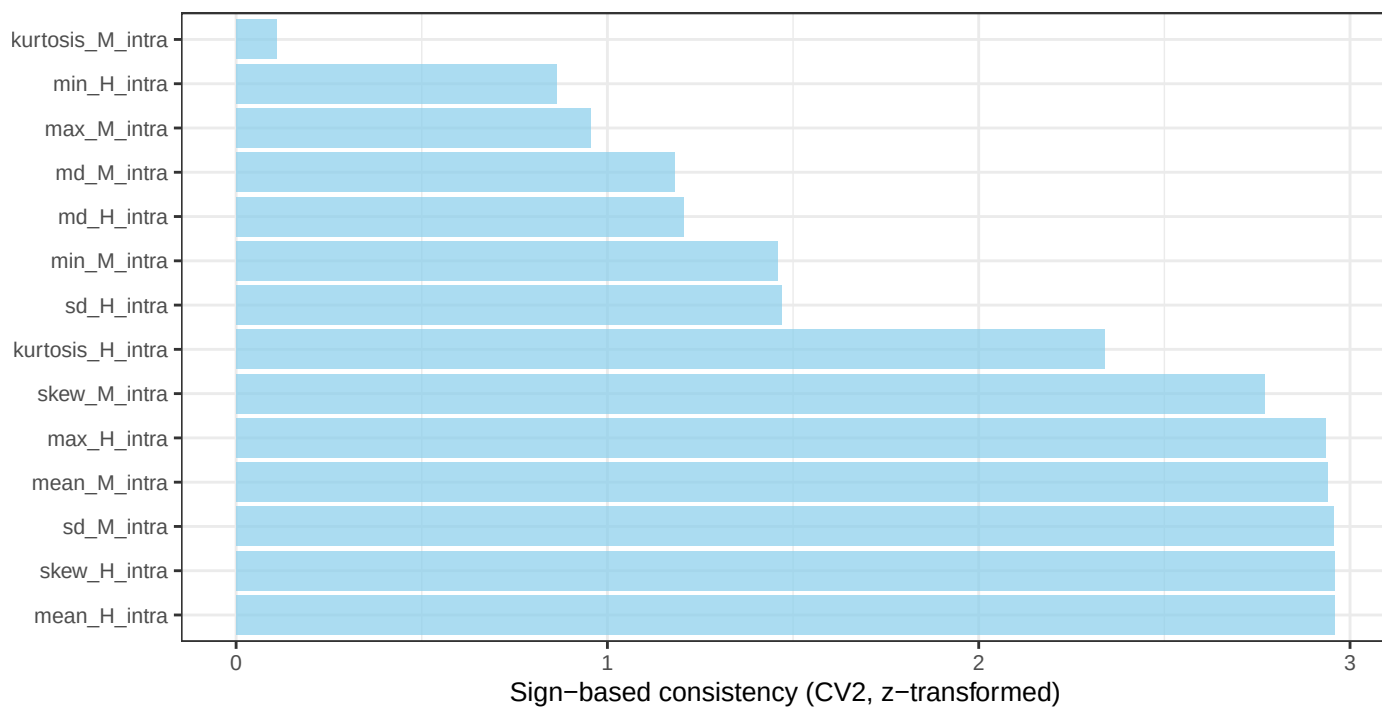
FACESync



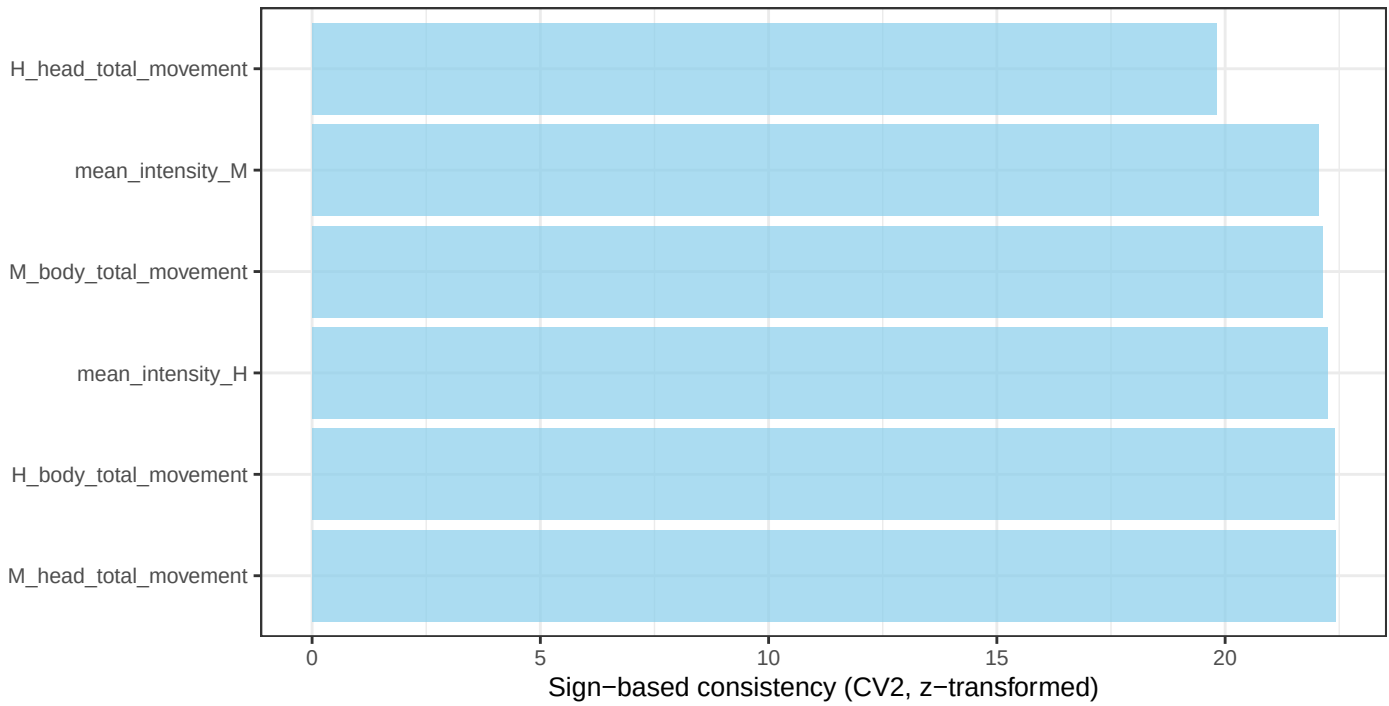
HEADsync



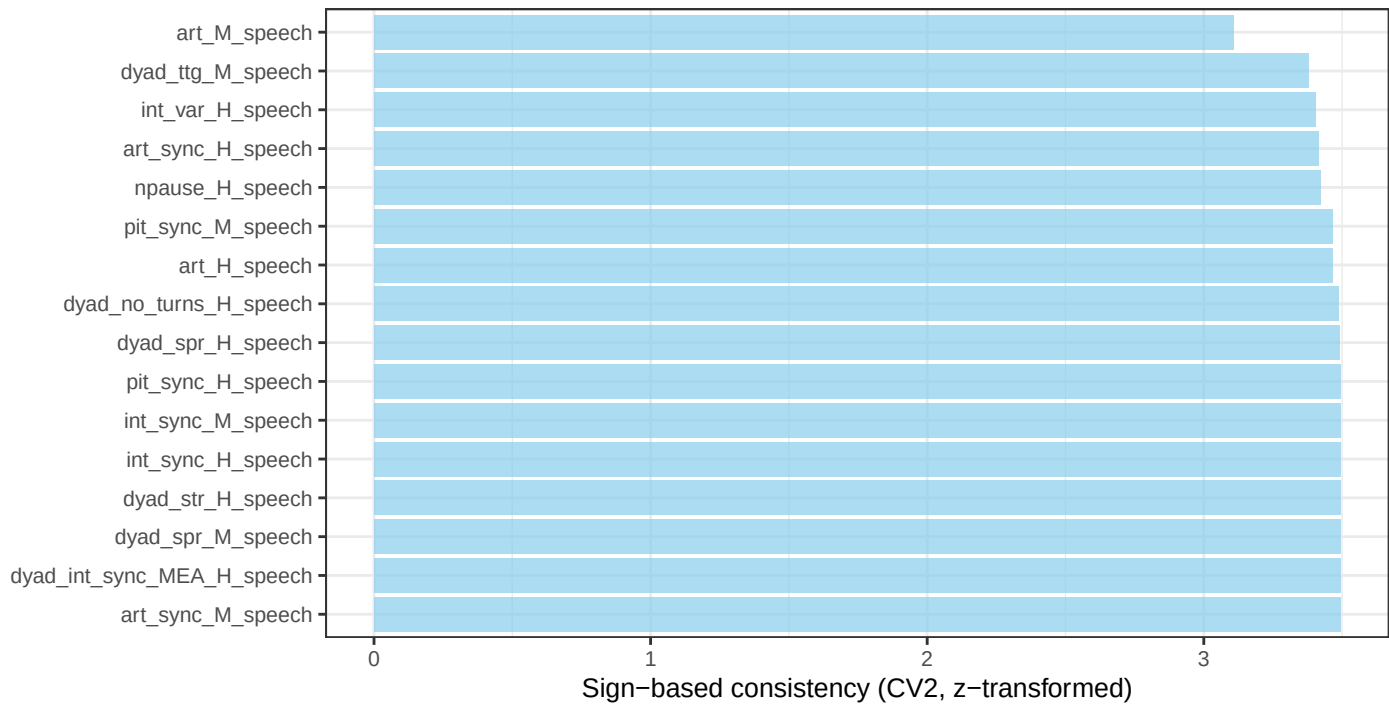
INTRAsync



MovEx

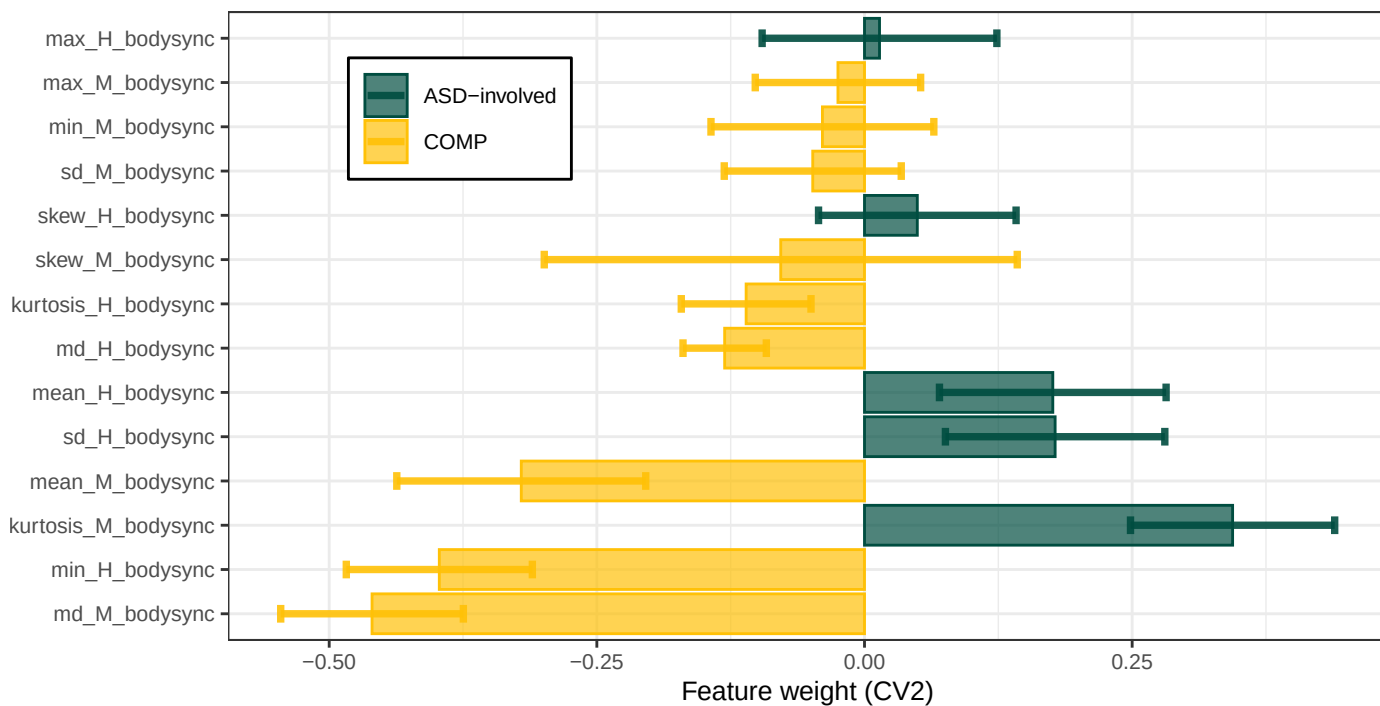


Speech

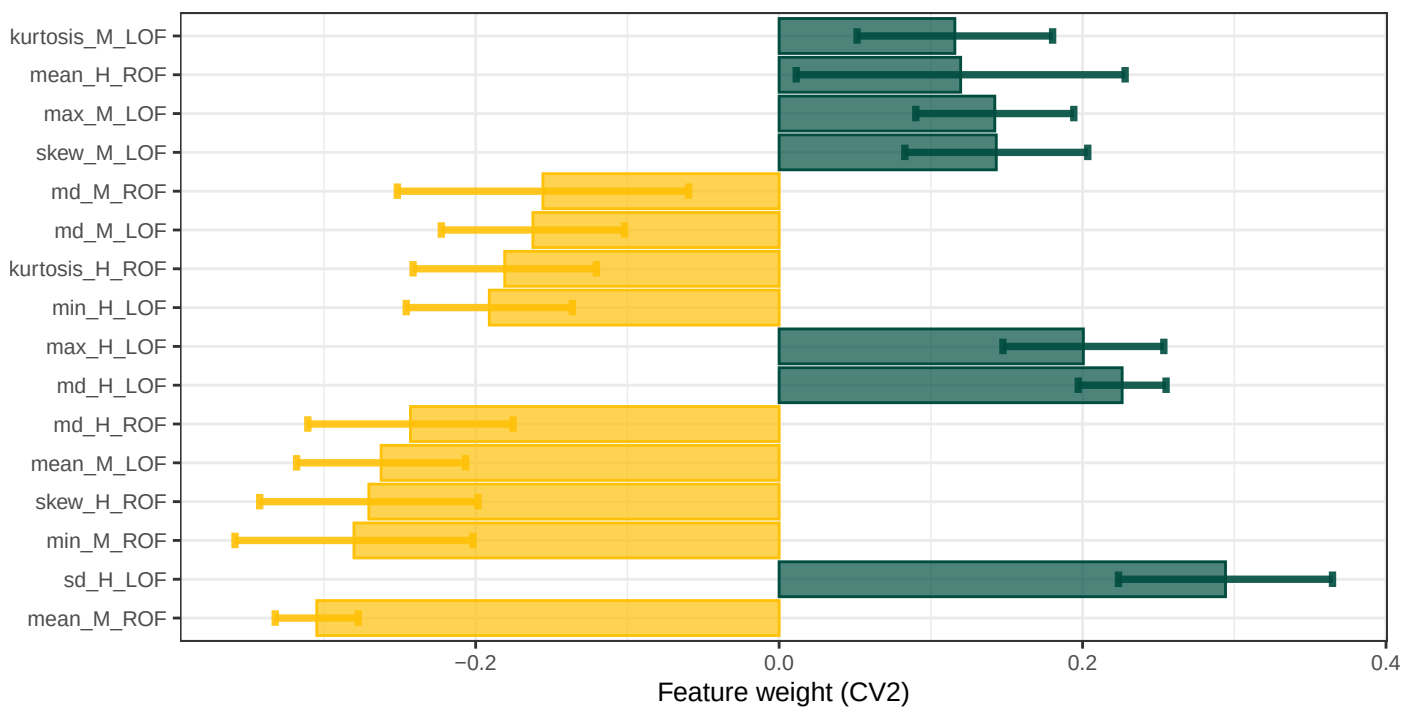


S10.2 ASD-involved versus COMP classifiers

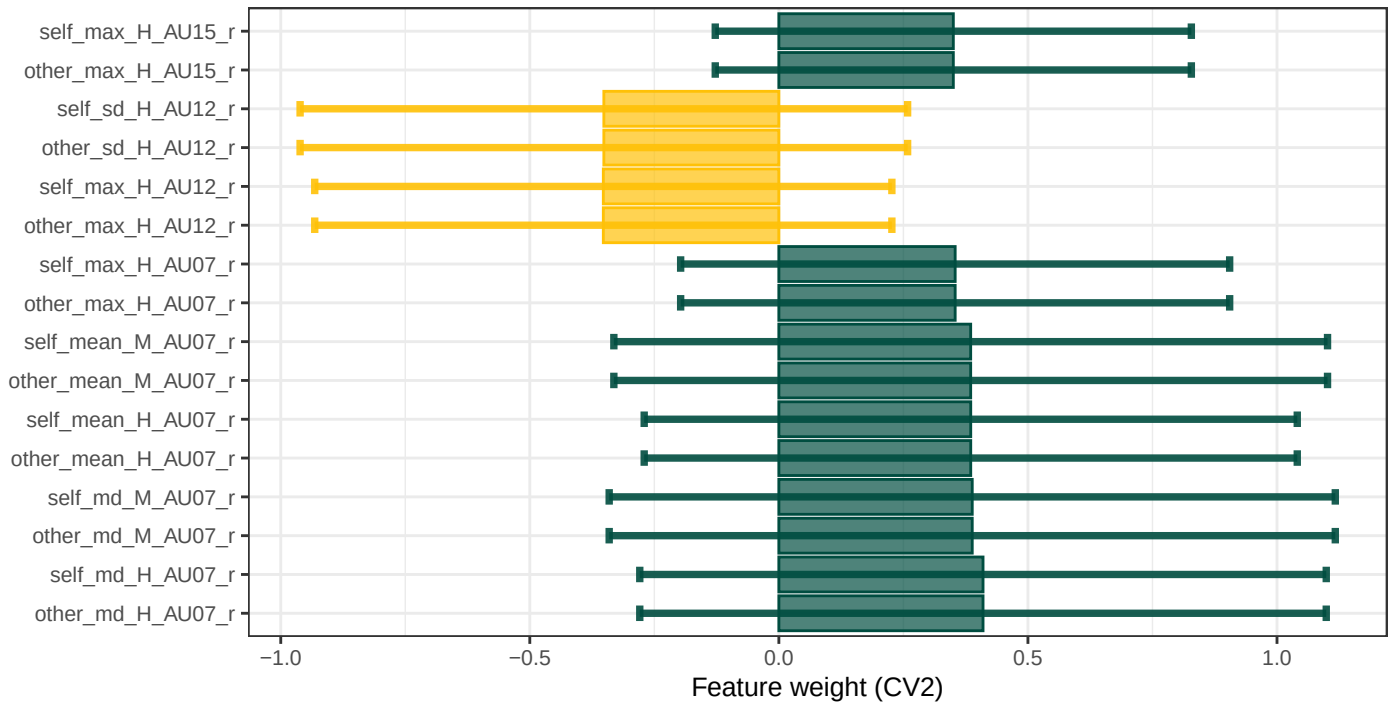
BODYsync



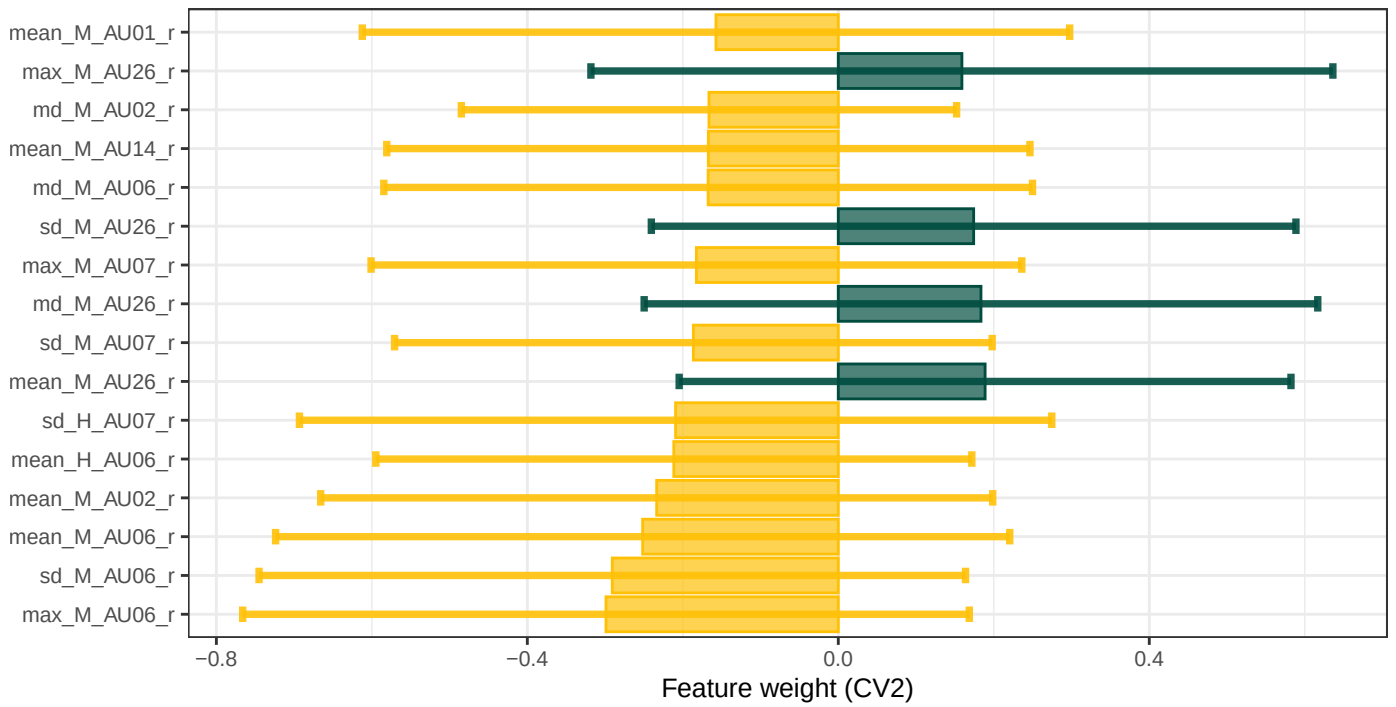
CROSSsync



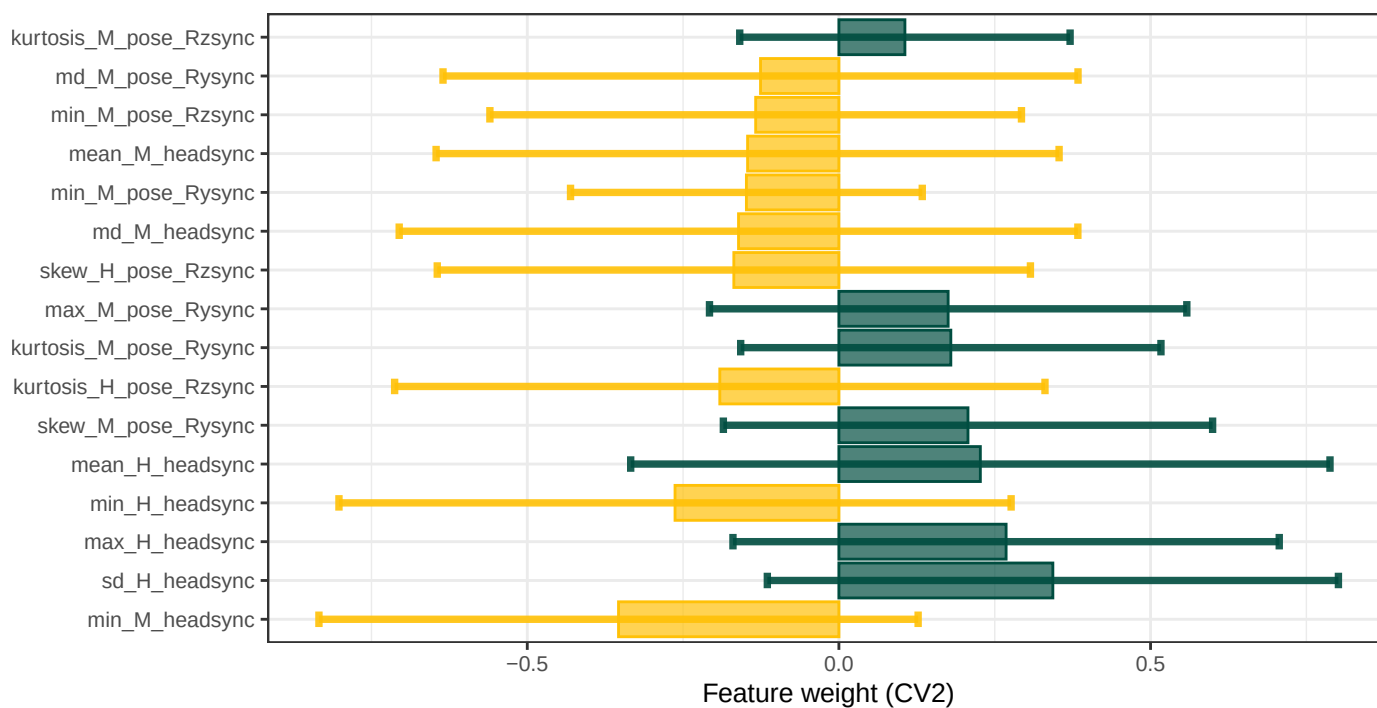
CROSSturn



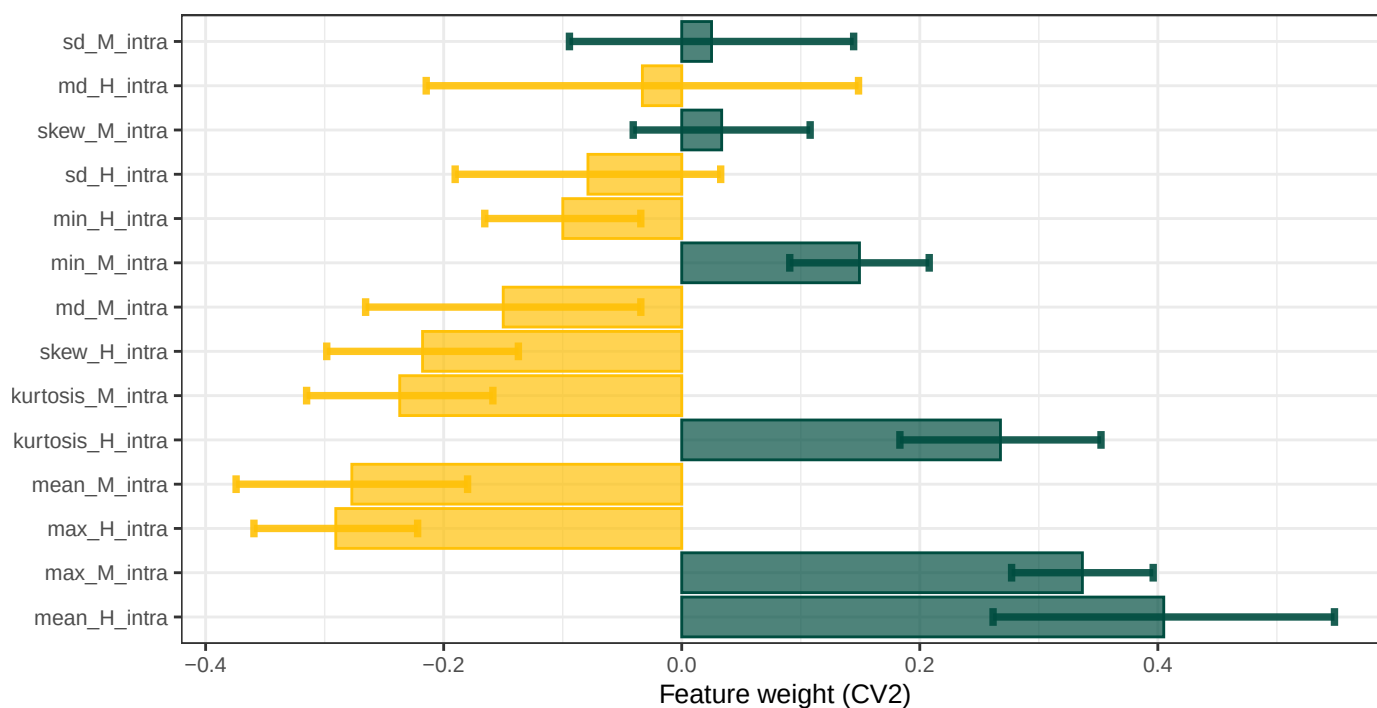
FACESync



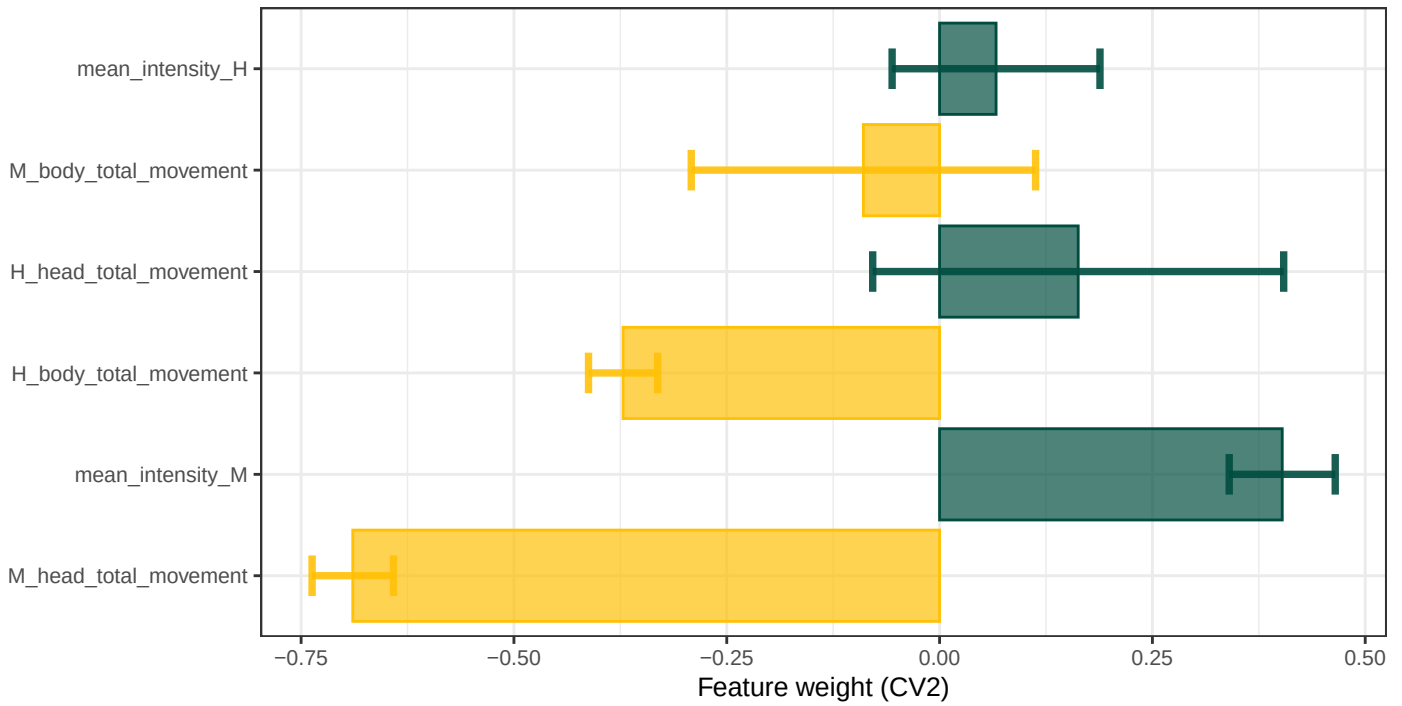
HEADsync



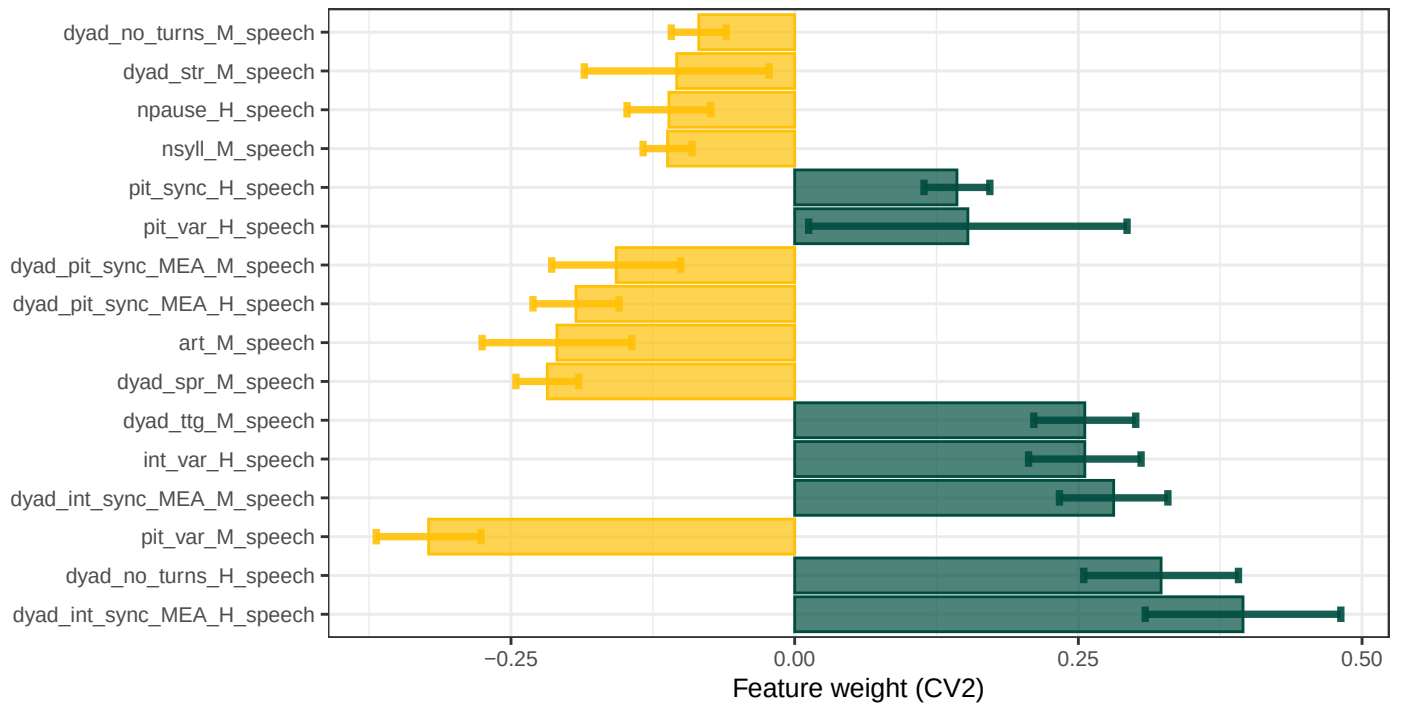
INTRAsync



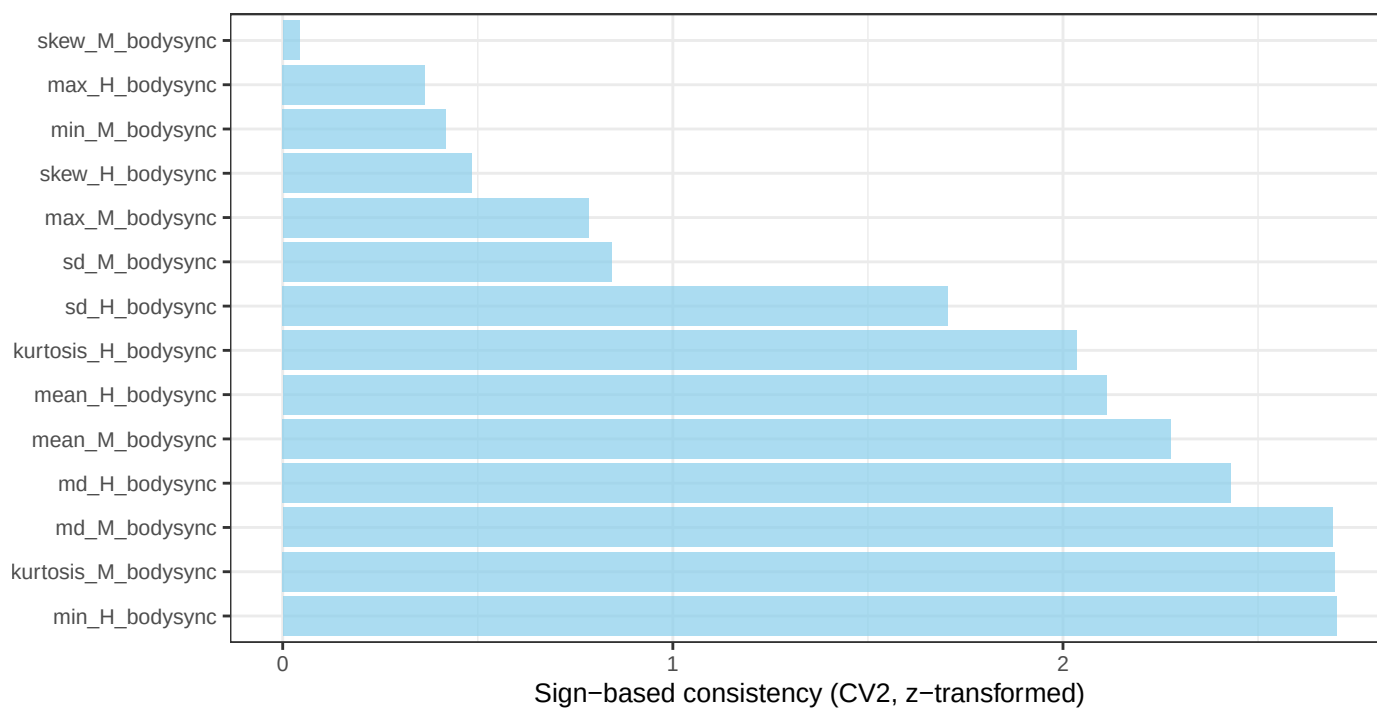
MovEx



Speech



BODYsync



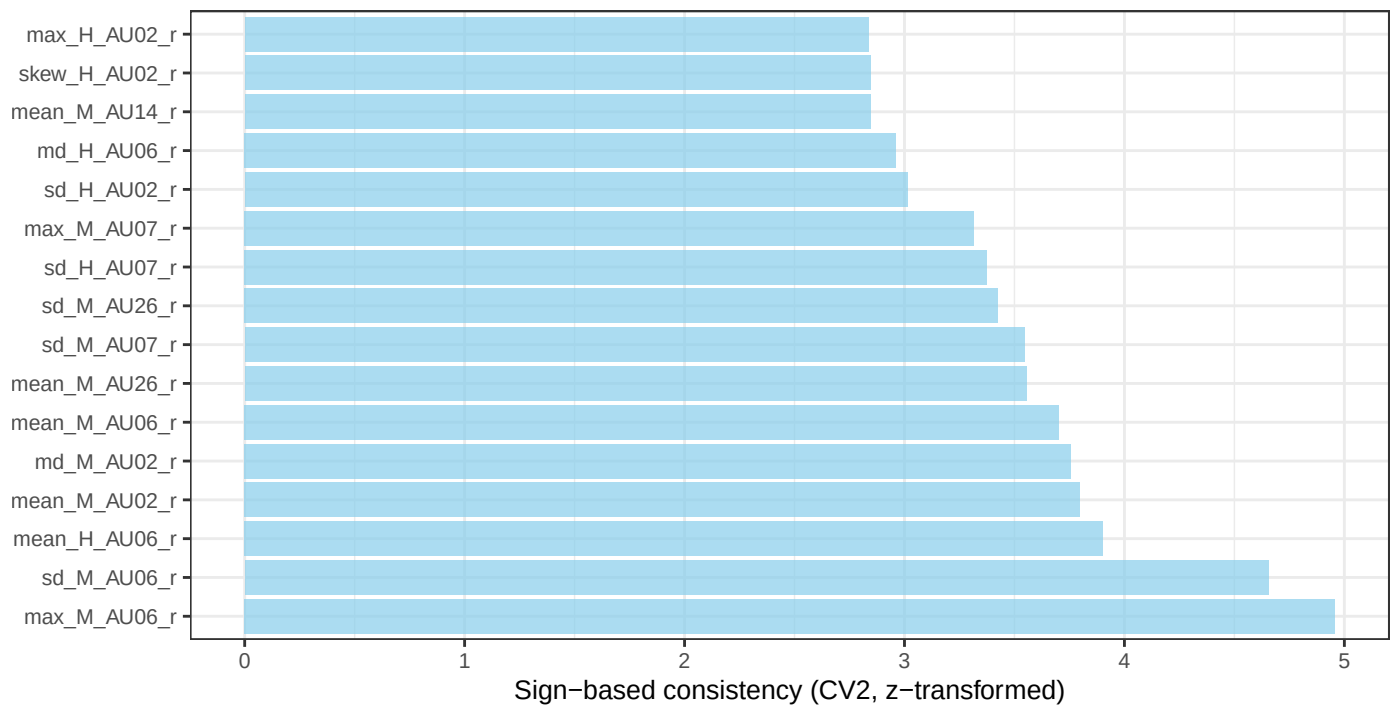
CROSSsync



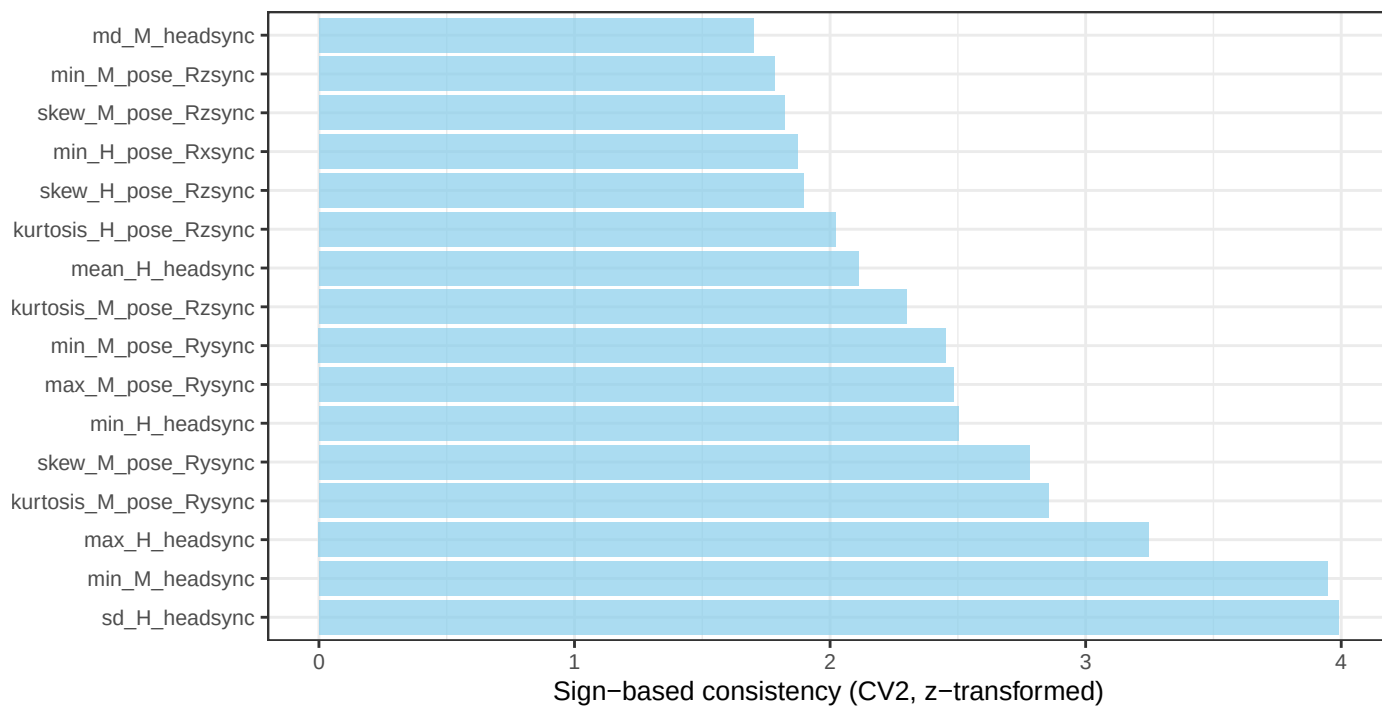
CROSSturn



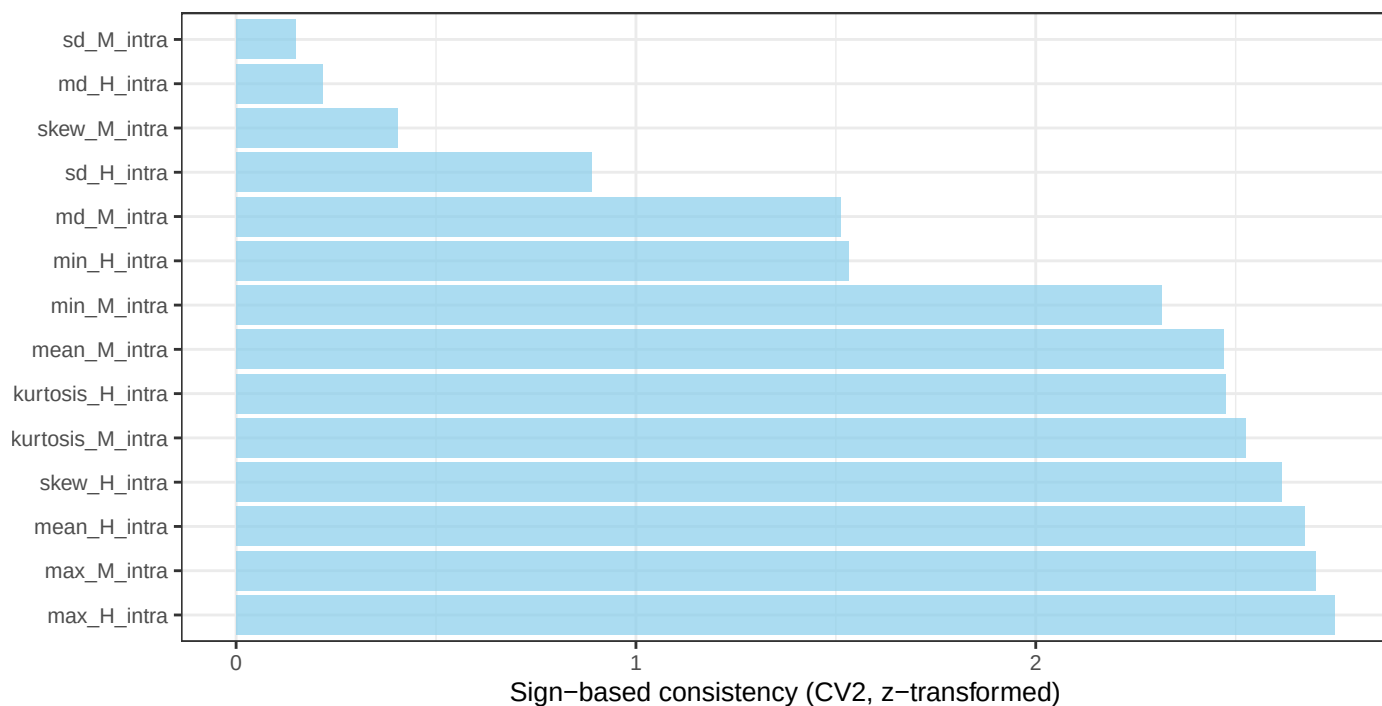
FACESync



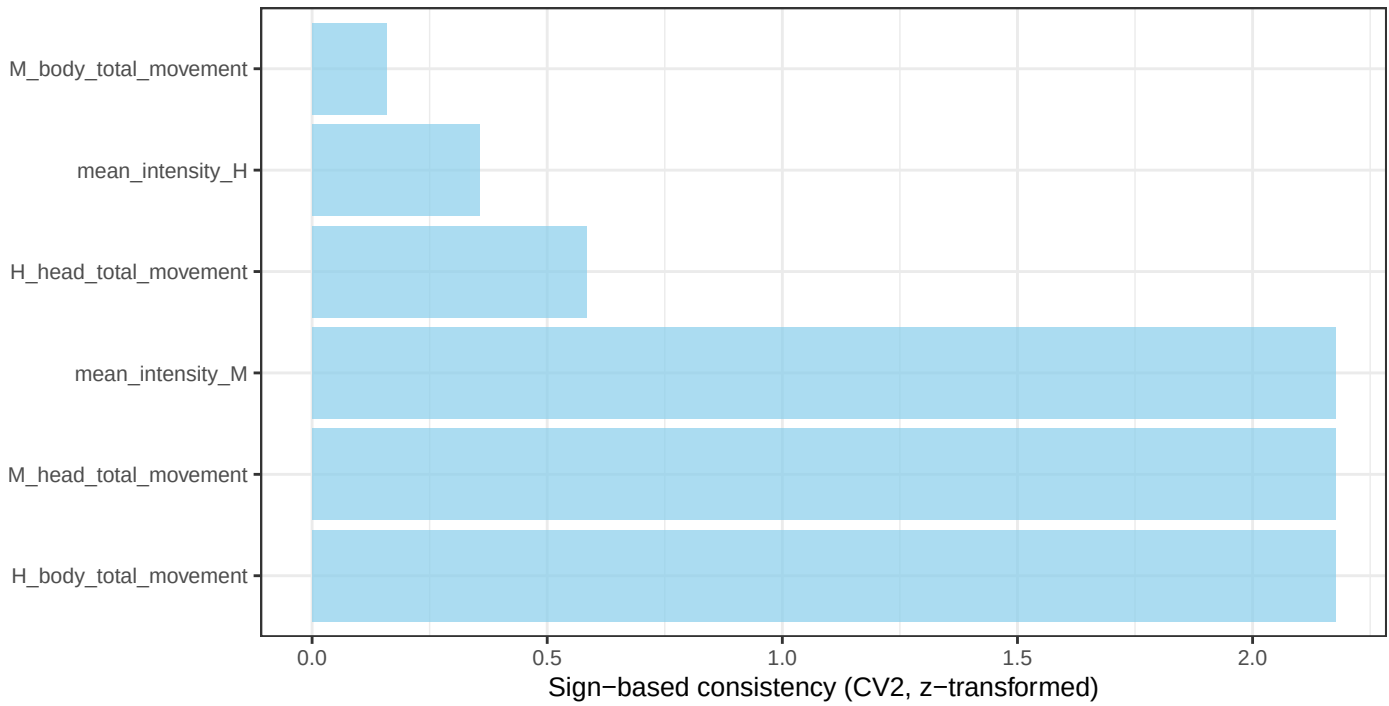
HEADsync



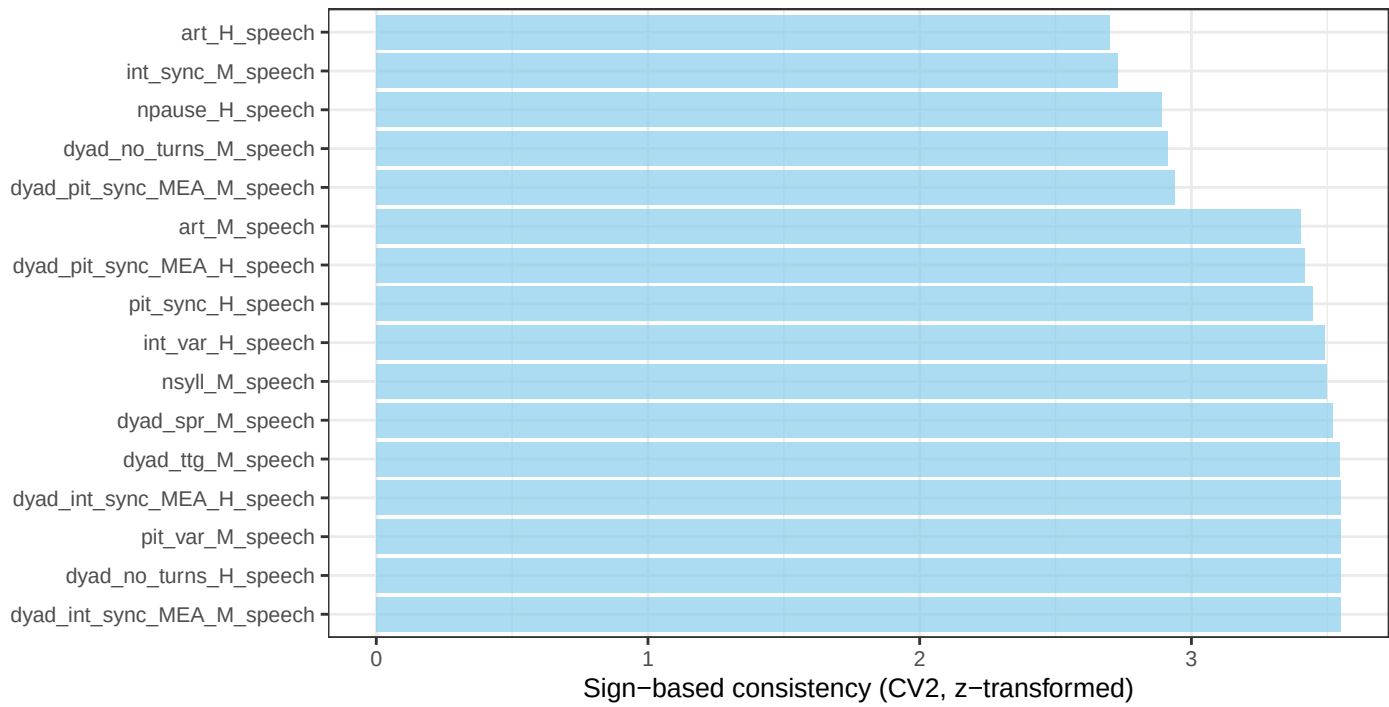
INTRASync



MovEx

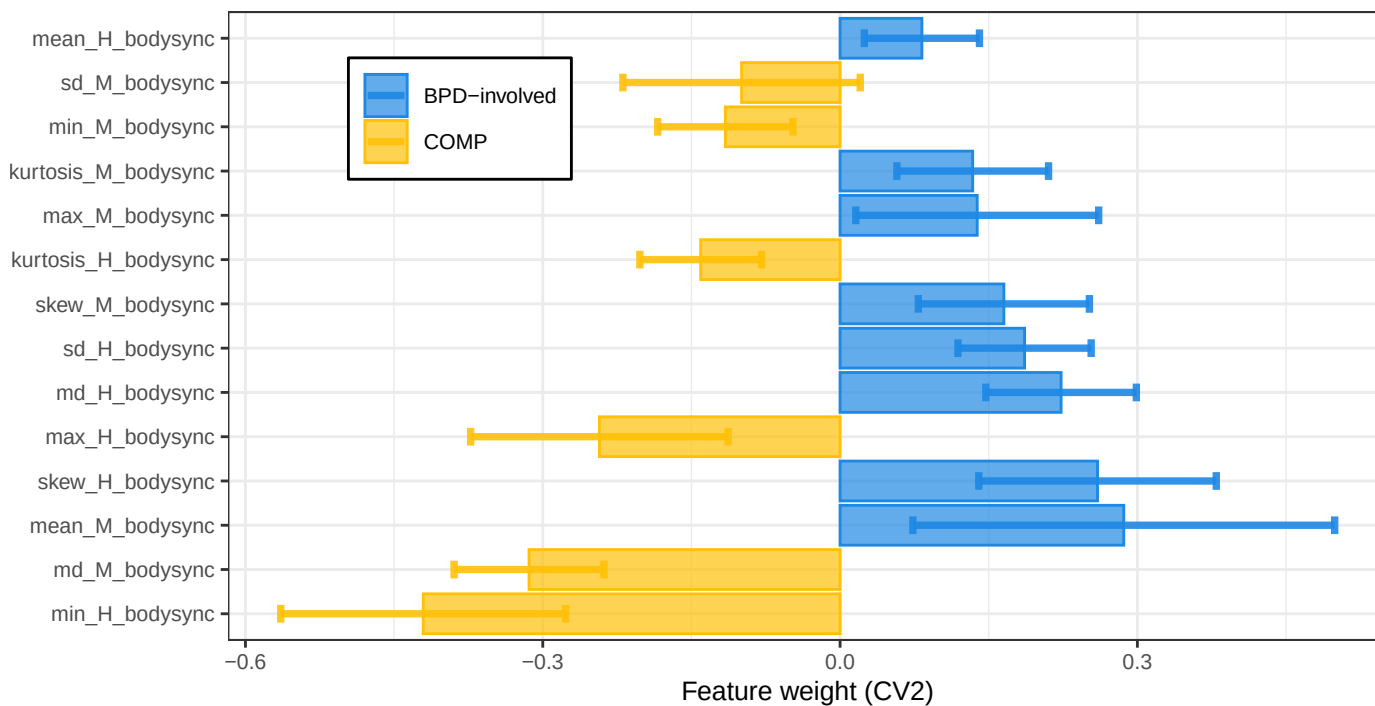


Speech

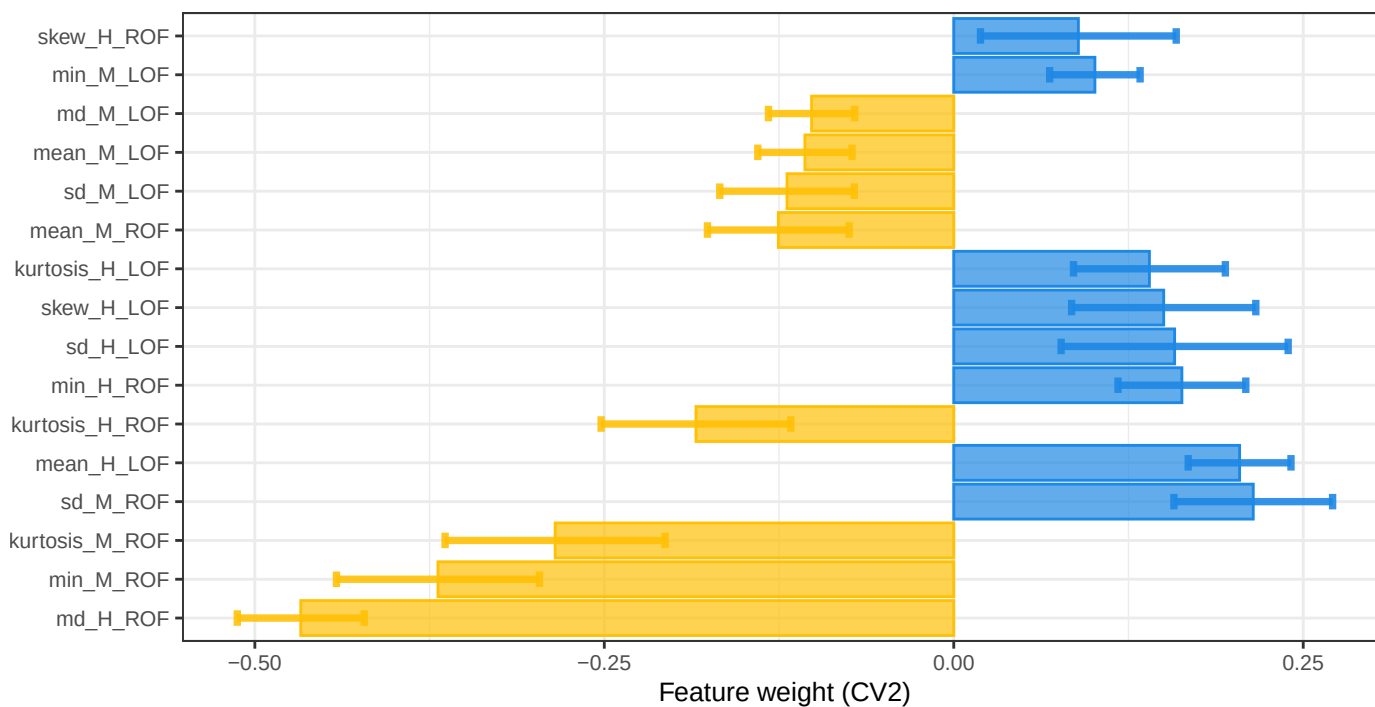


S10.3 ASD-involved versus COMP classifiers

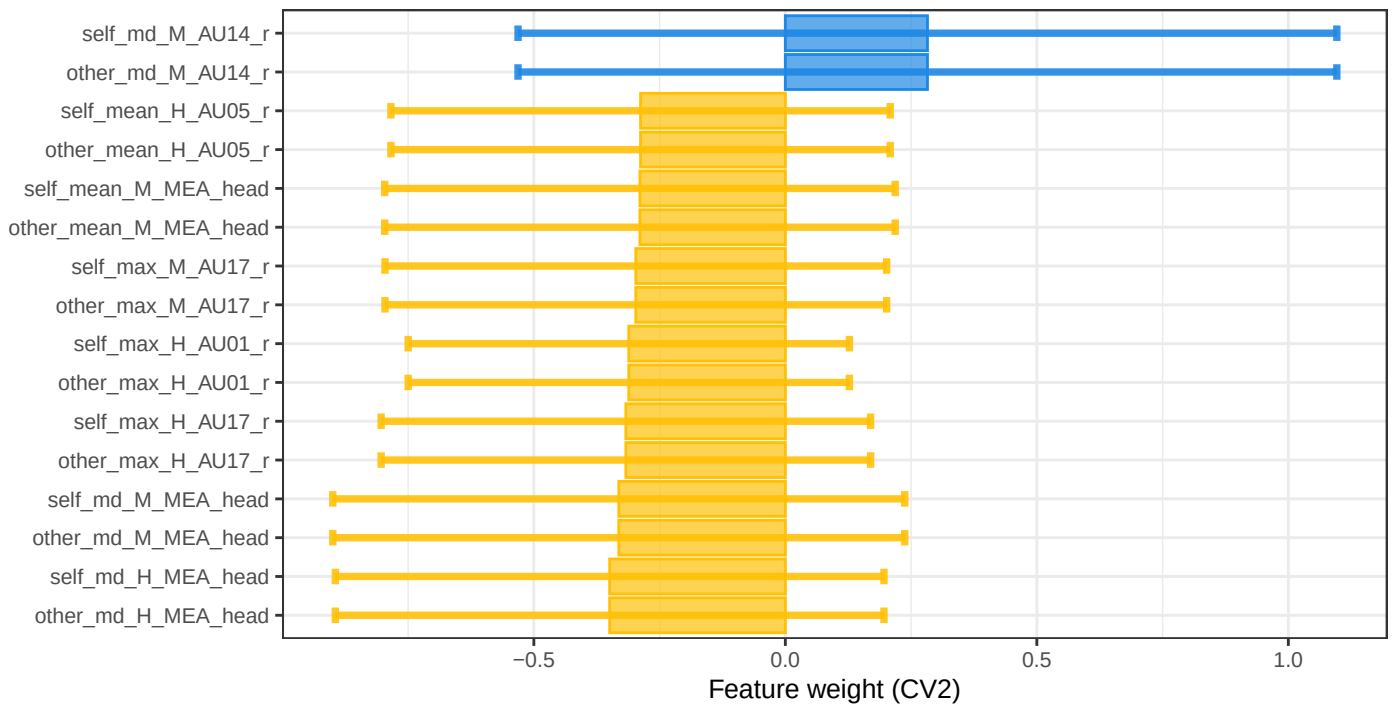
BODYsync



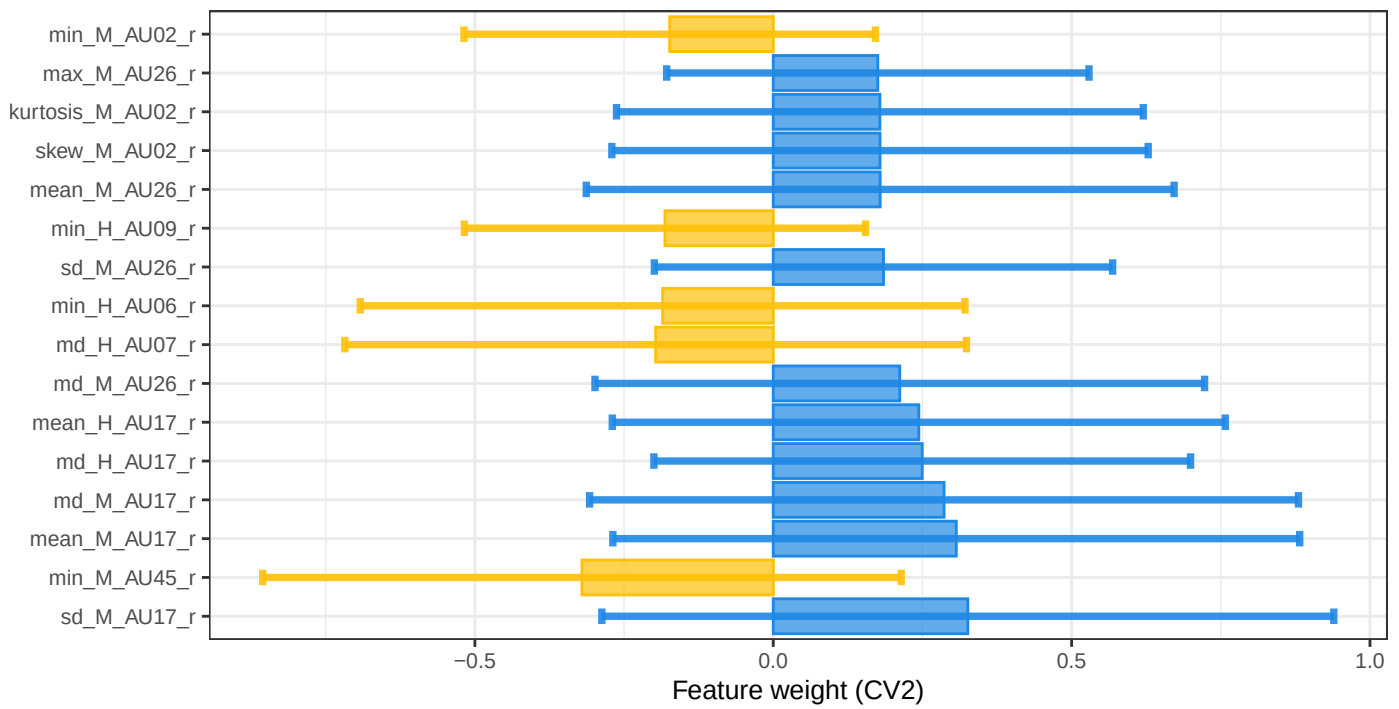
CROSSsync



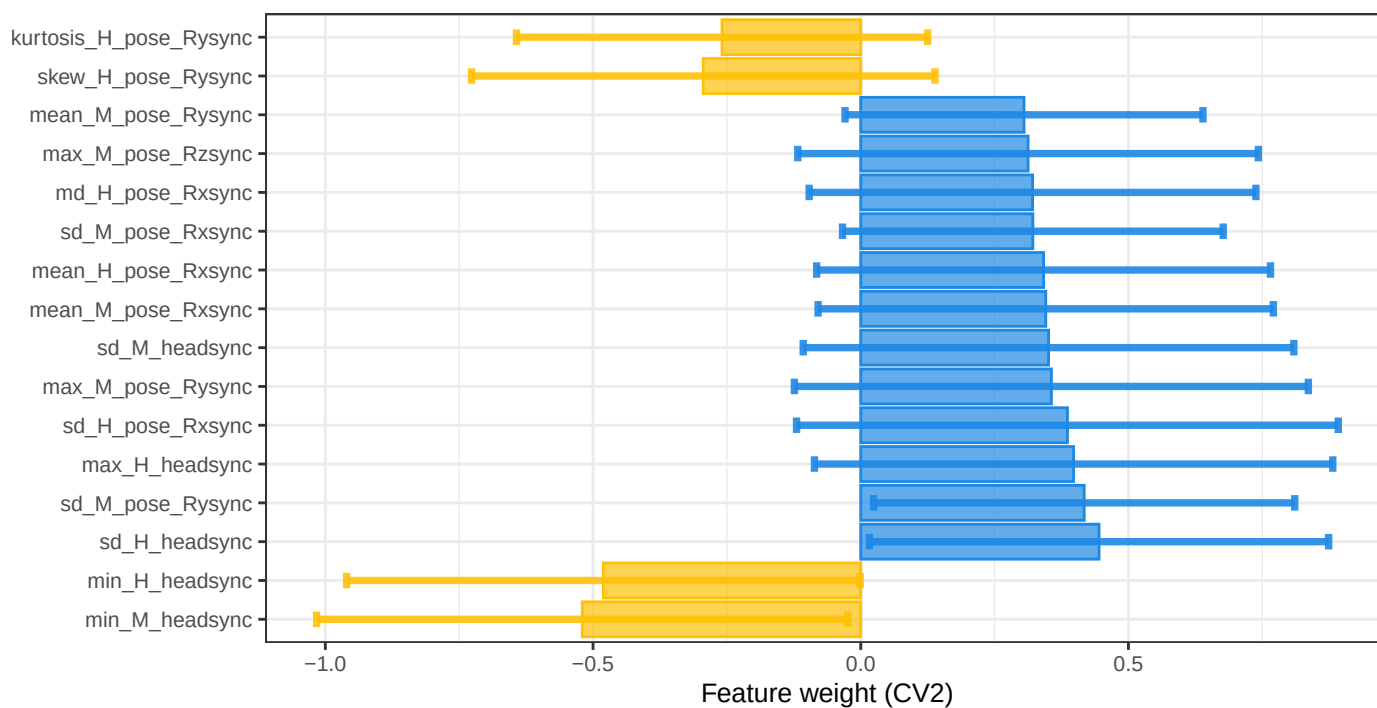
CROSSturn



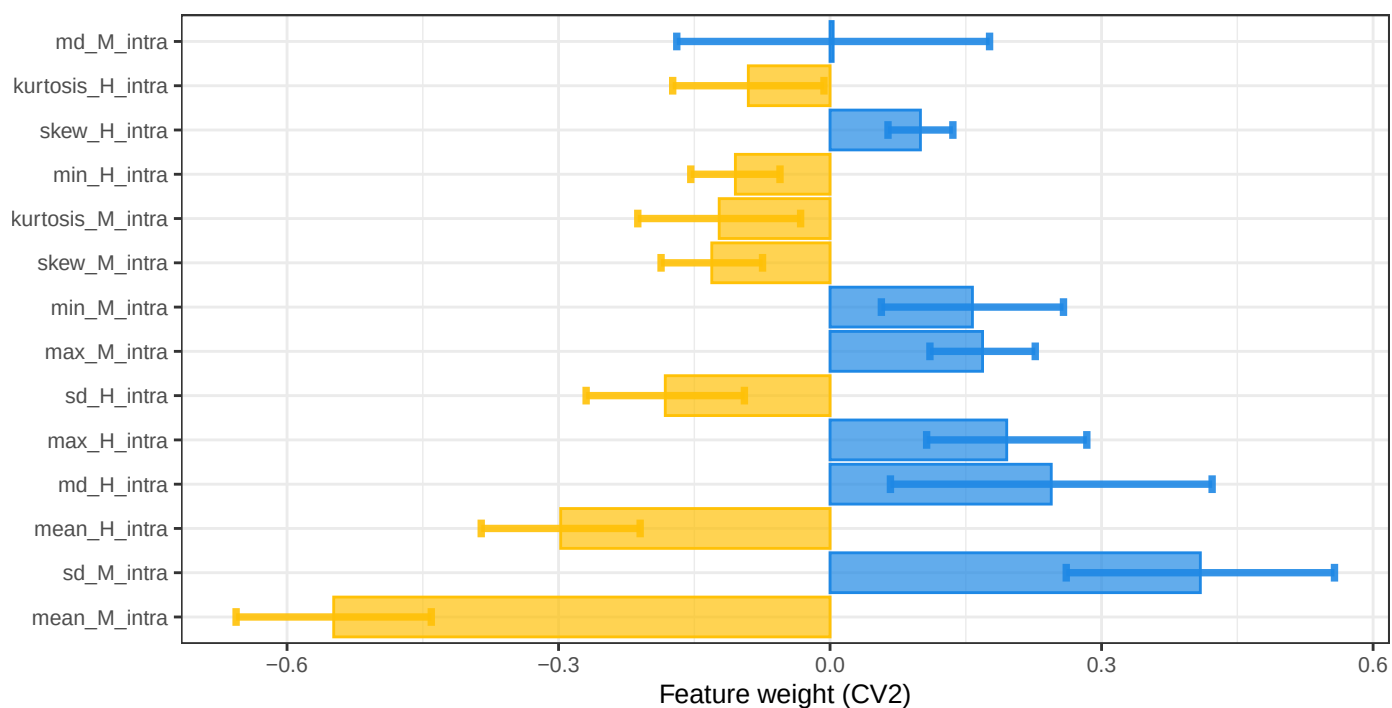
FACEsync



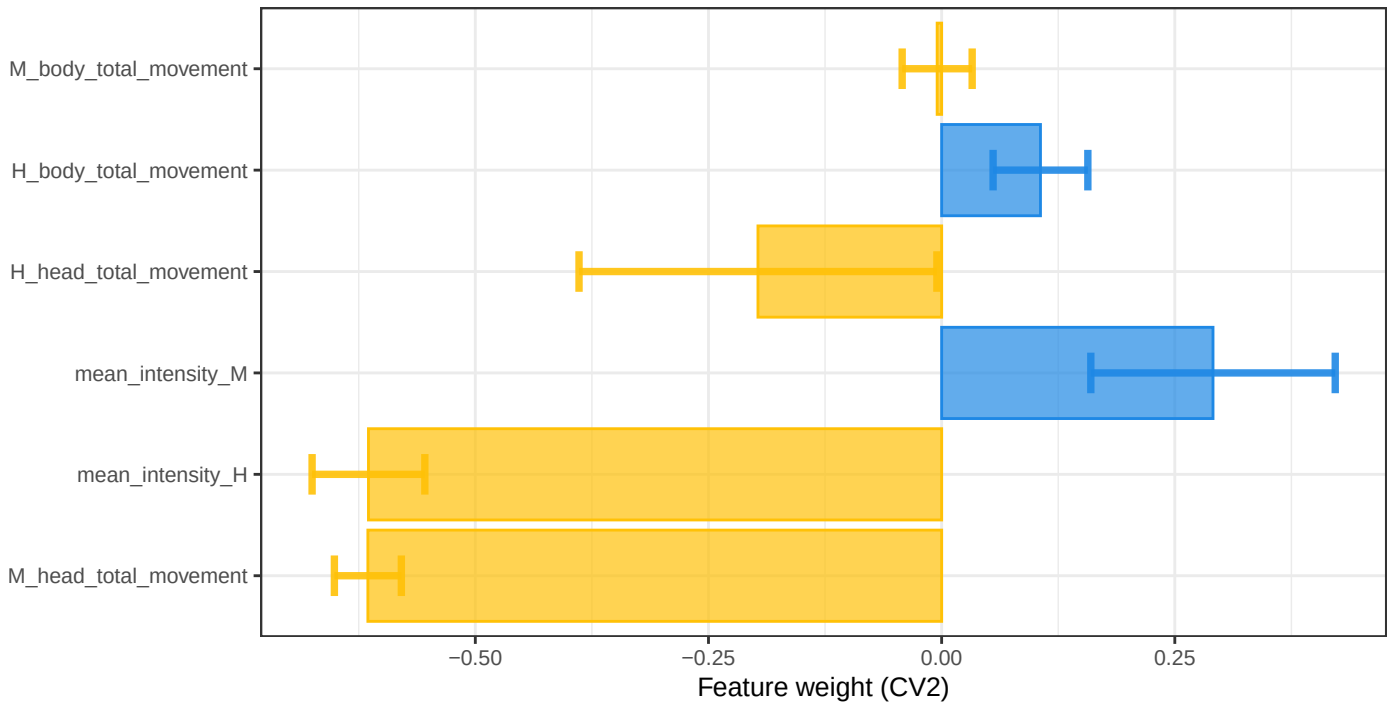
HEADsync



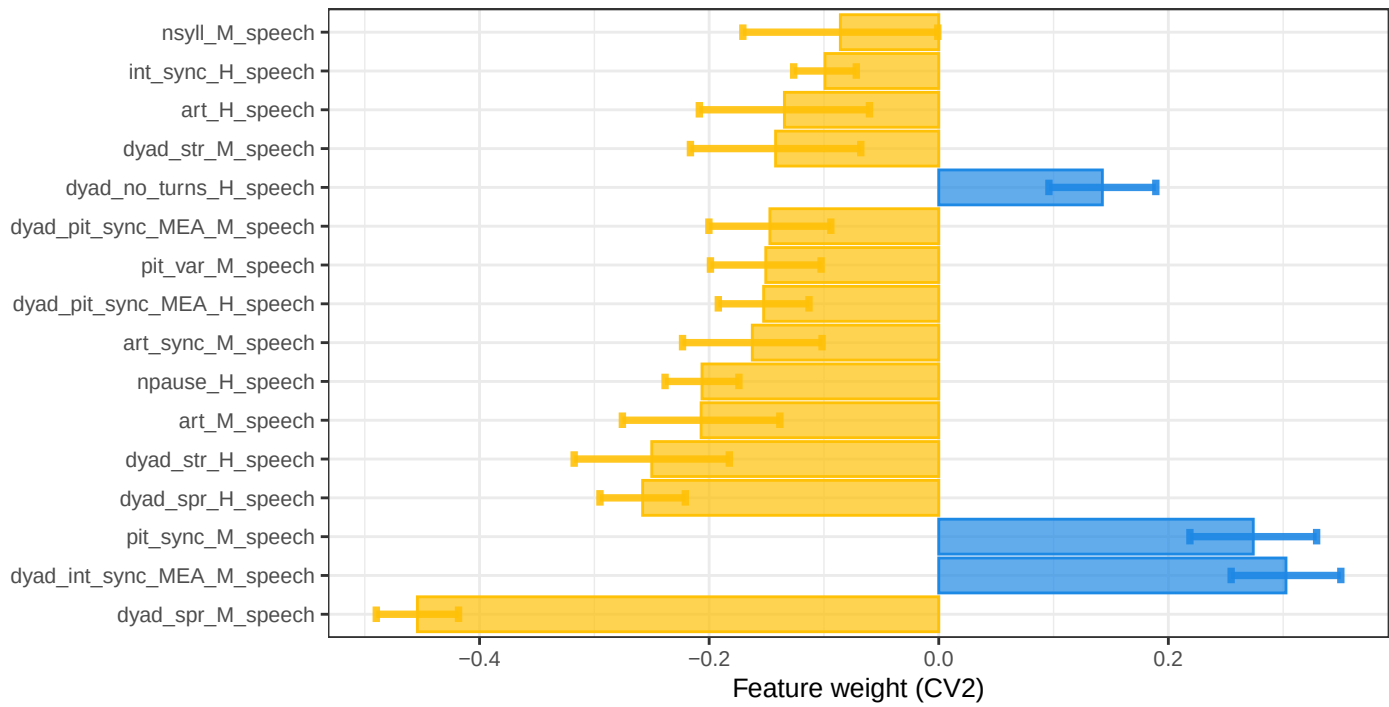
INTRAsync



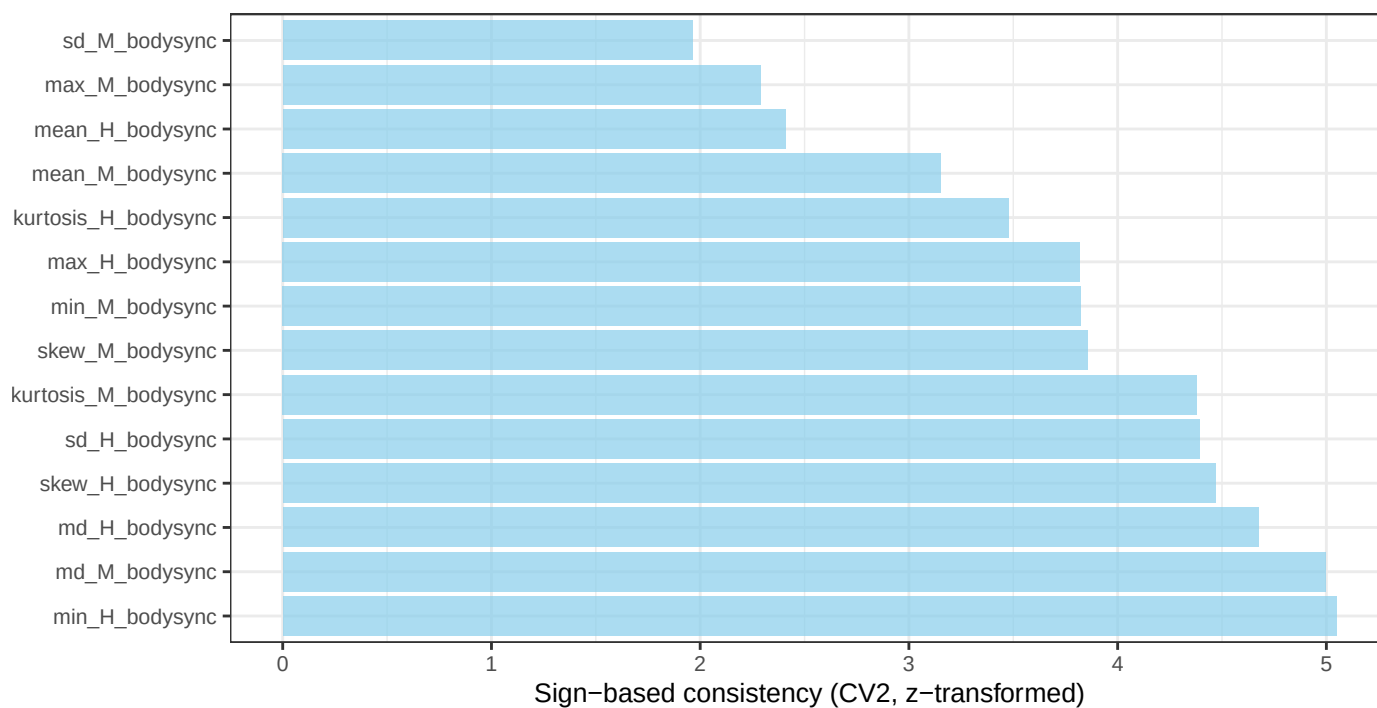
MovEx



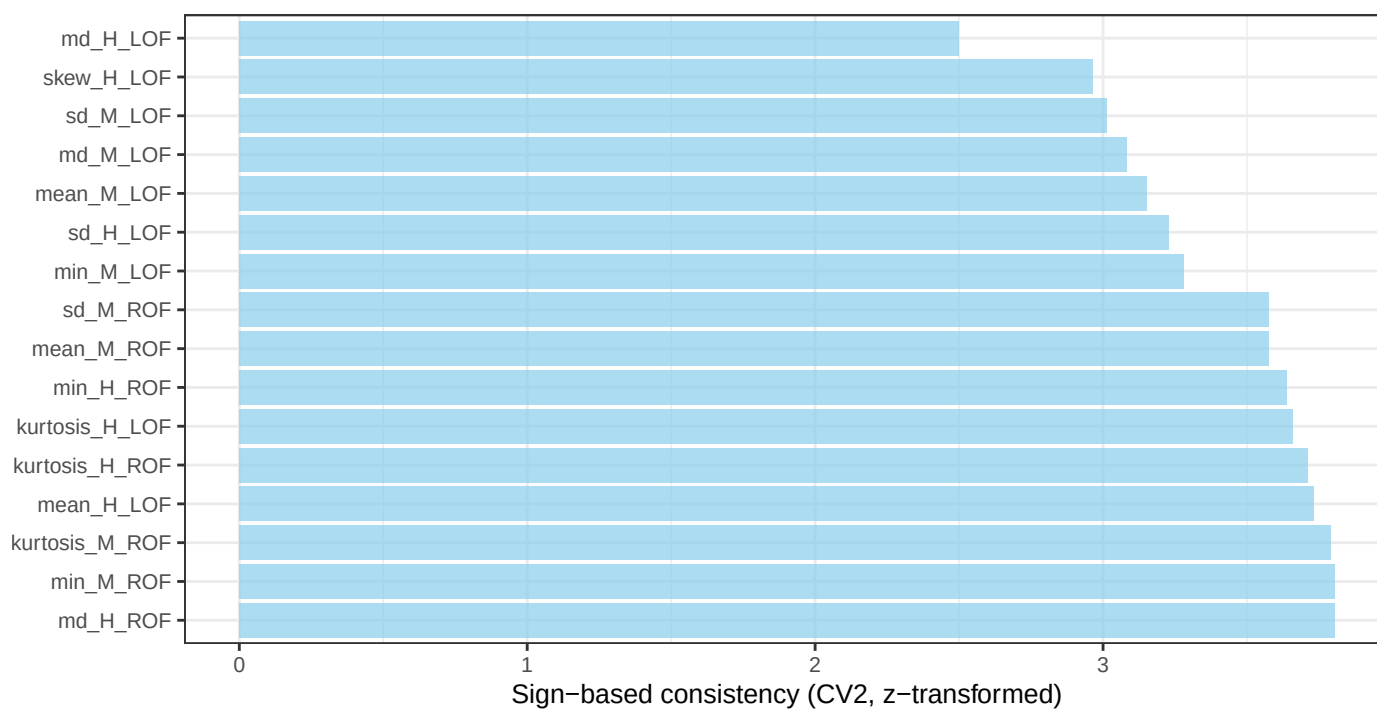
Speech



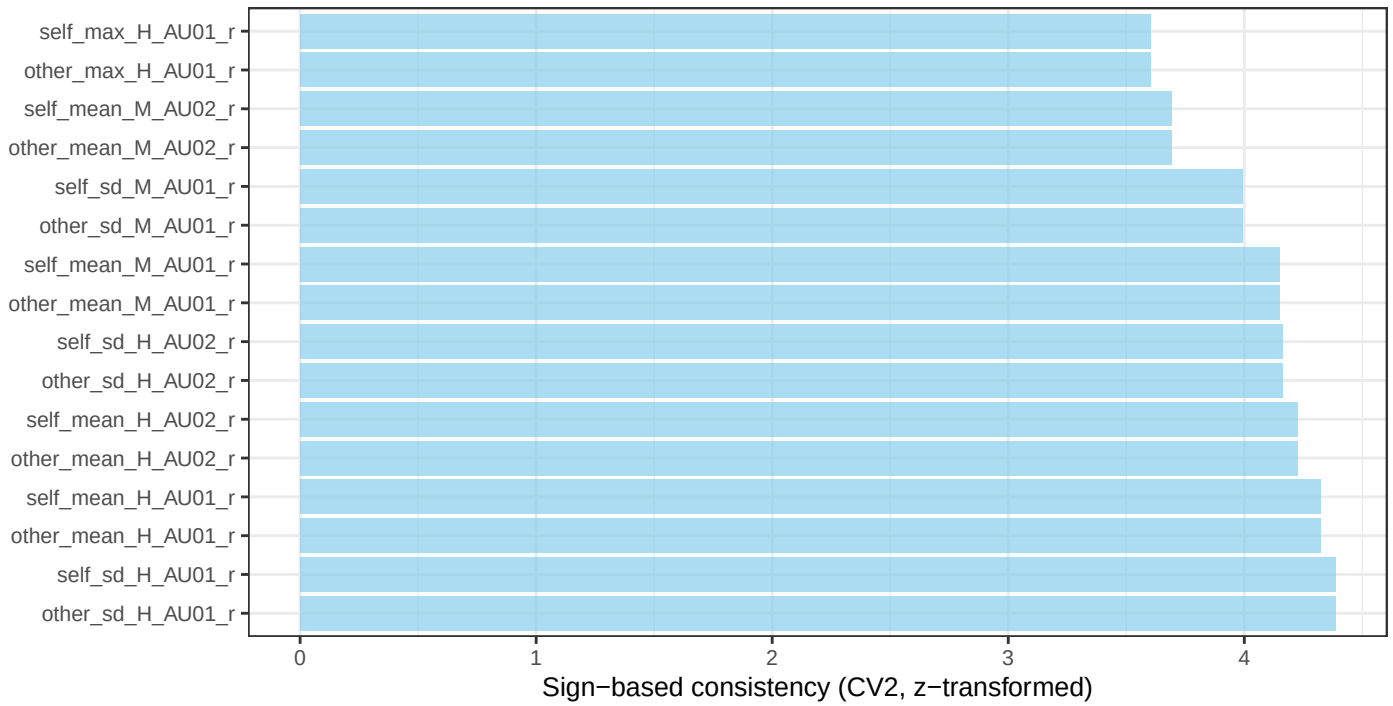
BODYsync



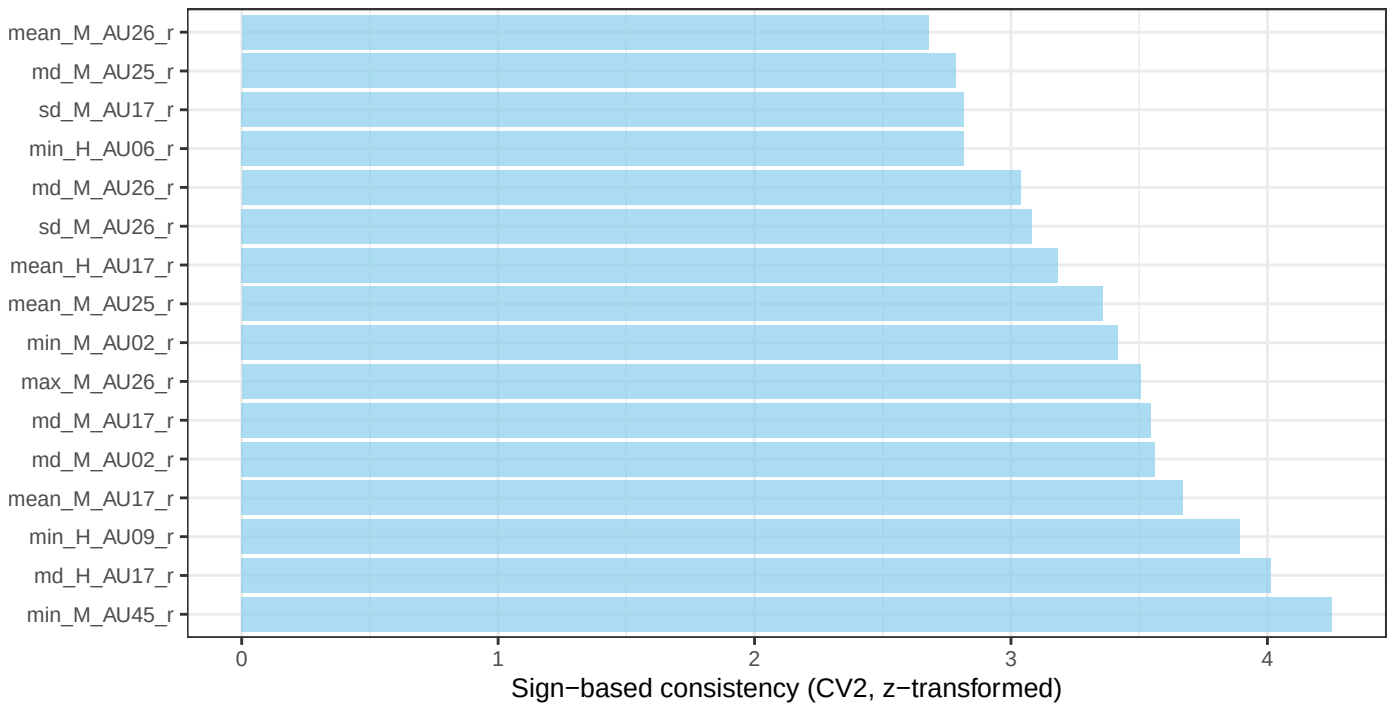
CROSSsync



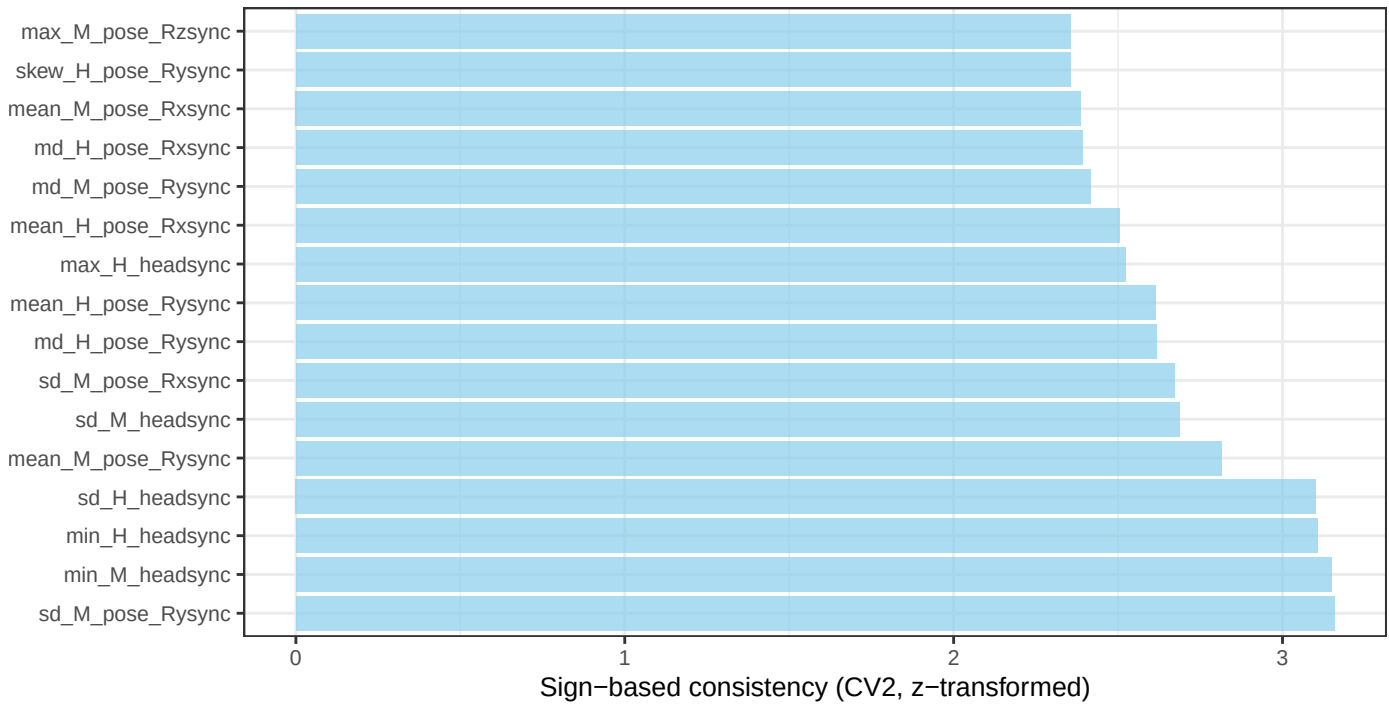
CROSSturn



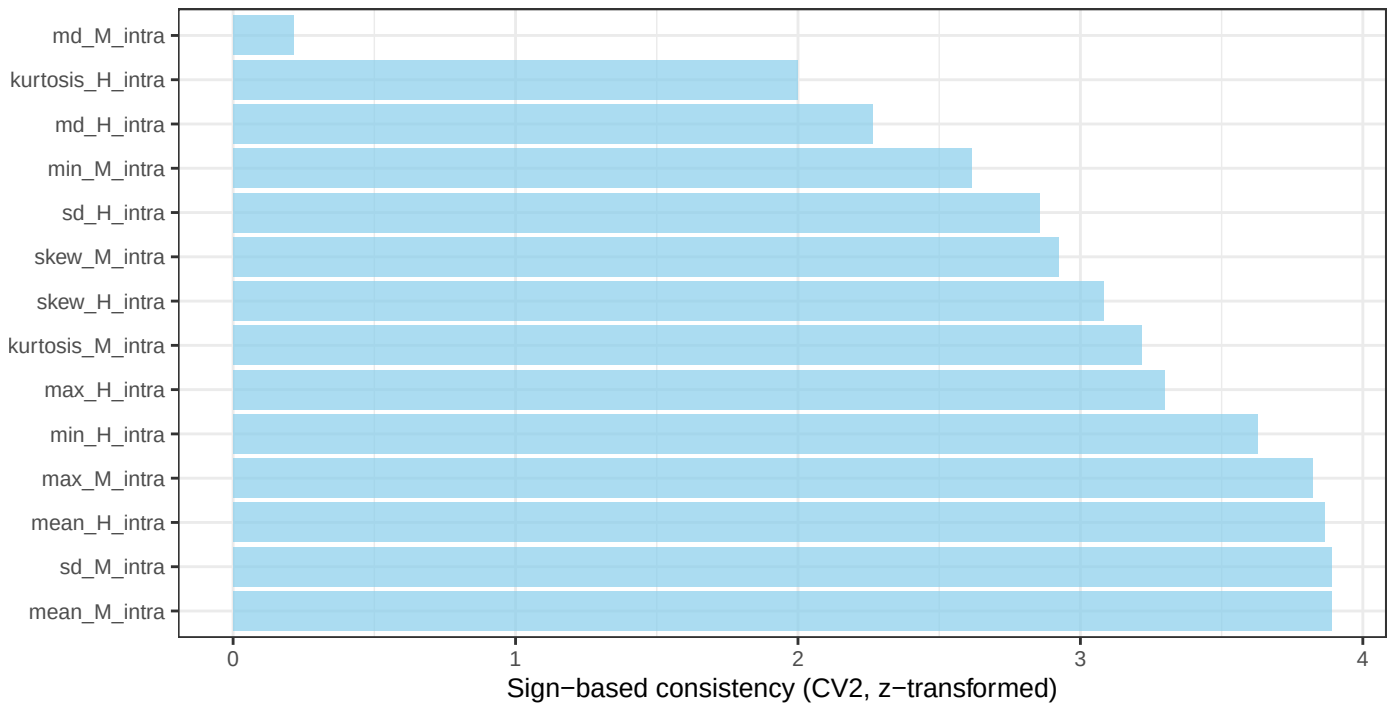
FACESync

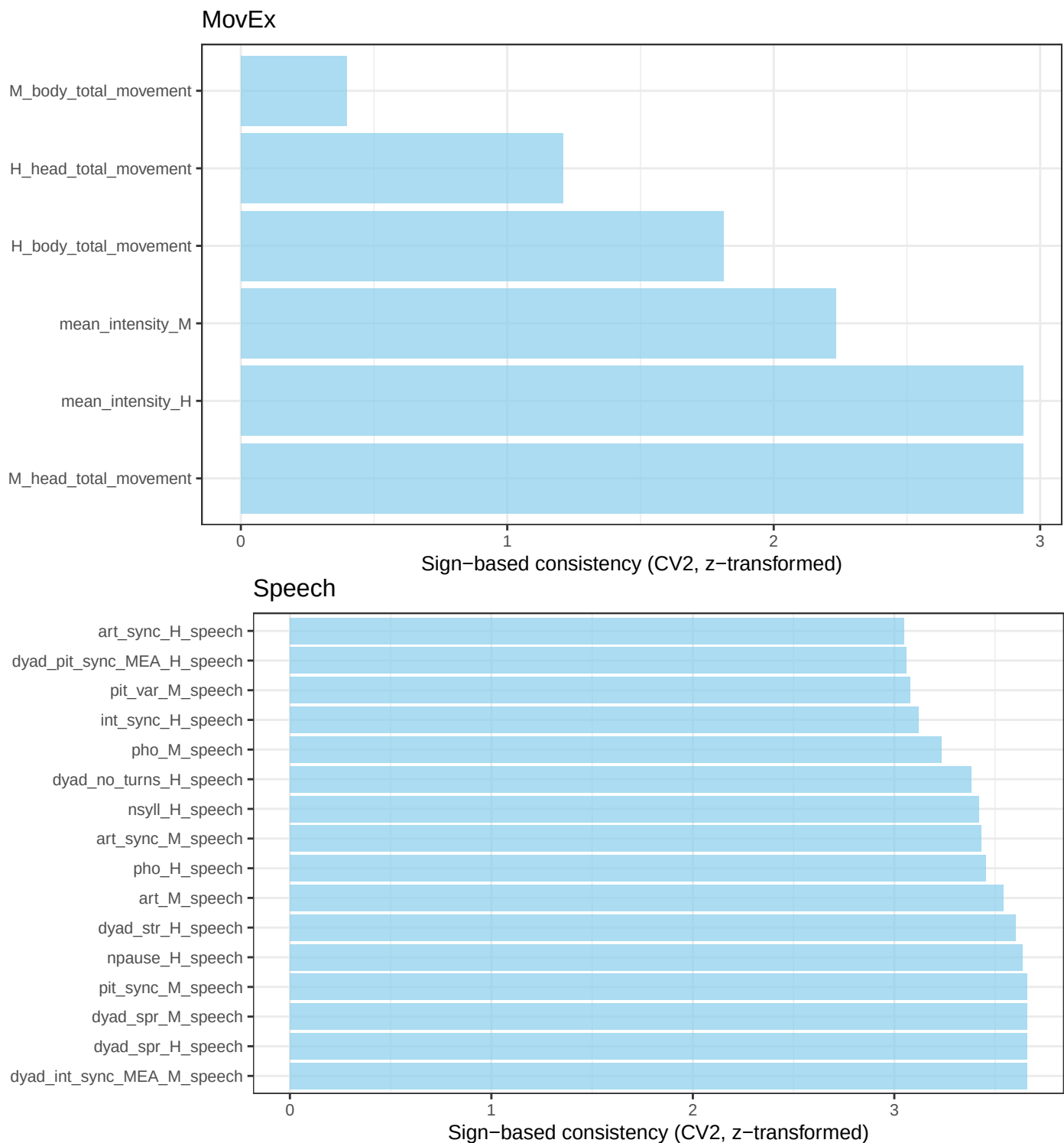


HEADsync



INTRAsync





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